

A multi-stage machine learning approach for stock price prediction: Engineered and derivative indices

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ABSTRACT

In this paper, a machine learning approach is proposed to predict the next day's stock prices. The methodology involves comprehensive data collection and feature generation, followed by predictions utilizing Multi-Layer Perceptron (MLP) networks. We selected 5,283 records of daily historical data, including open prices, close prices, highest prices, lowest prices, and trading volumes from four well-known stocks in the FTSE 100 index. A novel set of engineered and derivative indices is extracted from the original time series to enhance prediction accuracy. Two Multi-Layer Perceptron (MLP) are proposed to predict the next day's stock prices using the engineered discrete and continuous indices. The case study uses the daily historical time series of stock prices between January 1, 2000, and December 31, 2020. The proposed machine learning approach presents suitable applicability and accuracy, respectively.

1. Introduction

Stock price prediction is one of the most challenging problems in the field of financial time series due to its dynamism, nonlinearity, and chaotic nature, in which many parameters often change randomly (Kumar Chandar, Sumathi & Sivanandam, 2016; Ma, Han & Wang, 2021). Stock price prediction can be considered as an essential task for investors and financial institutions, because accurate prediction of the stock price can lead to suitable trade-off among return and risk of investment (Chen, Zhang, Mehlaawat & Jia, 2021; Tan, Yan & Zhu, 2019).

Given the inherent volatility of financial markets, advanced predictive models capable of adapting to ever-changing market conditions are urgently needed. Enhanced prediction accuracy can lead to improved investment decisions, ultimately benefiting both individual and institutional investors.

Portfolio optimization has turned into an important and challenging issue in the field of finance and investment. Portfolio optimization is a good combination of stocks such that the profit of one stock can offset the loss of other stocks (Mansini, Ogryczak & Speranza, 2015). Investors are usually interested in knowing the future status of the stocks they have invested in, and the measures that help them to form the best portfolio (Wang, Li, Zhang & Liu, 2020). Accordingly, stock selection

can be considered as a prerequisite and preliminary step for the portfolio optimization. Therefore, combination of portfolio optimization and forecasting theory in the form of using accurate models to predict the stock prices and trends can help the investors to pre-select the proper stocks with potential for greater profits in order to form an optimal portfolio (Ma et al., 2021; Yang, Chen, Li & Tang, 2019). Due to the importance of the stock market prediction, many researchers have addressed the issue including stock prices, stock returns, market indices and stock price trend.

Furthermore, while numerous studies have explored stock prices, few have focused on the potential of engineered and derivative indices to provide deeper insights into market dynamics and due to our best knowledge there is no certain study in which a complete set of derivative and engineered indices are used to predict the stock price time series rather than direct forecasting of data.

The main purpose of this paper is to develop a machine learning approach to predict the next day's stock prices using artificial neural networks (ANNs) through a novel and comprehensive set of engineered and derivative indices. A key innovation of this study lies in the extraction of derivative indices from original stock price time series data. By focusing on these engineered indices, we aim to enhance the predictive capabilities of our model. Instead of relying solely on raw

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Table 1
Selected stock prediction literature.

Researcher	Dataset	Feature set		Applied Method
		Input Variables	Target	
Ma et al. (2021)	49 stocks from China Securities 100 Index	Past 60 days' daily returns	Next day's return	DMLP, LSTM neural network and CNN, random forest, support vector regression
Ribeiro et al. (2021)	Three Nasdaq companies: Caterpillar (CAT), eBay (EBAY), and Microsoft (MSFT).	Past values of series	Realized volatility of stock prices	Echo state network Heterogeneous autoregressive model Particle swarm optimization
Kumar Chandar (2021)	SBIN.NS, NTPC.NS, TATASTEEL.NS, INFY.NS	Opening price, highest price, lowest price and closing price, several technical indicators	Intraday stock price	Nine hybrid model based on ANN
Zhang et al. (2020)	SHCOMP, SZSE, GEM, HIS, DJIA, S&P 500 Index	Extracted Intrinsic Mode Functions (IMFs)	One-step-ahead closing price	LSTM
Ismail et al. (2020)	Kuala Lumpur Composite Index (KLCI)	Input vectors of invariant topological features	Next day direction of stock price movement	LR, ANN, SVM, and RF
Vijh et al. (2020)	Nike, Goldman Sachs, Johnson and Johnson, Pfizer and JP Morgan Chase and Co.	H-L, O—C, 7 DAYS MA, 14 DAYS MA, 21 DAYS MA, 7 DAYS STD DEV	next day closing price	ANN, Random Forest
Tan et al. (2019)	Chinese stock market	Fundamental/Technical variables as well as momentum	long-term and short-term stock price trend	Random forest (RF)
Yang et al. (2019)	China's A-share stock market	Close, open, high, low, average price, MCAP, return rate, volume, turnover, turnover rate, volatility, general capital, P/E, P/B, PC/F, and P/S.	Price, Return	Extreme learning machine
Bisoi et al. (2019)	BSE S&P 500 Index, Hang Seng Index, and Financial Times Stock Exchange 100 Index	Lagged intrinsic mode functions (IMFs) and lagged modes	Price, Trend	Robust kernel extreme learning machine
Paiva et al. (2019)	São Paulo Stock Exchange Index	Return lagged based on opening, closing, maximum, and minimum prices as well as several technical variables such as Momentum, RSI, ATR, Chaikin A/D line, on balance volume	Determine the stock has the potential to gain 1% return in $t + 1$	SVM
Zhou et al. (2019)	NASDAQ, S&P 500	Extracted Intrinsic mode functions (IMFs) from Stock Close Price.	Determining the changes (directions) and predicting the future prices (trends) in stock prices.	Neural Networks
Basak et al. (2019)	AAPL, AMS, AMZN, FB, and MSFT	RSI, SO, %R, MACD, PROC, OBV	Direction of stock market prices	random forests, and gradient boosted decision trees
Lee and Yoo (2018)	10 stocks in the S&P500	open, high, low, close, and volume	Return	RNN, GRU, LSTM
Lei (2018)	SSE Composite Index, CSI 300 Index, All Ordinaries Index, Nikkei 225 Index, and Dow Jones Index	Random % K, Random % D, W&R, MTM, RSI, BBI,5BIAS, 10BIAS, MB, PSY, AR, BR, MV, TAPI,VR	stock price trend	Wavelet Neural Network, Rough Set
Hiransha, Gopalakrishnan, Menon and Soman (2018)	National Stock Exchange of India, and New York Stock Exchange	Daily closing price	stock price	MLP, RNN, LSTM, and CNN
Chen and Hao (2018)	Shanghai and Shenzhen stock exchange markets	Open, high, low, and close prices. Technical indicators such as: MA, EMA, MACD, RSI, OBV, MoM, ATP ¹³ , ALT ¹⁴ , ITL ¹⁵	Stock trading signal classification (considering four level for target including strongly recommended buying signal (SBS), ordinarily recommended buying signal (OBS), ordinarily recommended selling signal (OSS), and strongly recommended selling signal (SSS). One-month-ahead stock returns	Weighted SVM
Abe and Nakayama (2018)	Japanese stock market	Fundamental Features	Labeling each stocks according to their ranking with respect to the return-to-volatility ratio	Logistic Regression, RF, DNN
Fu et al. (2018)	Chinese stock data	Technical and Fundamental Features	Stock Market One-Day Ahead Movement: 1: Open ($i + 1$) – Close(i) 2: Open ($i + 1$) – Open(i) 3: Close ($i + 1$) – Close(i) 4: Close ($i + 1$) – Open(i) 5: Trade Volume ($i + 1$) – Trade Volume(i)	ANN, DT, SVM
Weng, Ahmed and Megahed (2017)	AAPL, NASDAQ	Open, high, low, and close prices, and volume of trades, RSI ¹⁰ , P/E ¹⁶ , %K ¹⁷ , %R ¹⁸	Trend, Peak Detection (PK), Trough Detection (TRU), Two Consecutive Peaks (DBP), Distance Between Two Consecutive Troughs (DBT), Distance from One Peak or Trough to The Next (DBPT), Peak Height	Multi-Layer Perceptron (MLP)
This Study	Randomly chooses four stocks from FTSE 100	Several lagged variables of returns, several technical indicators		

(continued on next page)

Table 1 (continued)

Researcher	Dataset	Feature set		Applied Method
		Input Variables	Target	
			(PKHe), Trough Depth (TRUDE), Distance Between, Delta, Rate of Return (Ret.)	

P/E: Price-Earnings Ratio, P/B: Price-to-Book Ratio, PC/F: Price-to-Cash-Flow Ratio, P/S: Price-to-Sales Ratio, H-L: Difference between High price and Low price, O—C: Difference between Open price and Close price, MA: Moving Average, STD DEV: Standard Deviation, MCAP: Market Capitalization, RSI: Relative Strength Index, ATR: Average True Range, SO: Stochastic Oscillator, %R: Larry William's R%, %K: Stochastic %K, %D: Stochastic %D, MACD: Moving average convergence divergence, PROC: Price rate of change, OBV: On balance volume, EMA: Exponential Moving Average, MoM: Momentum Index, ATP: Average Transaction Price, ALT: Amplitude of the Price Movement, ITL: Index for the Type of K-line, A/D Oscillator: Accumulation/Distribution Oscillator.

price data, we will predict stock prices using this set of derivative indices rather than applying original stock price time series, thereby uncovering deeper insights into market dynamics. Most existing models focus on direct forecasting using raw price data, neglecting the potential insights that can be gained from engineered features. This research seeks to uncover underlying patterns and trends that traditional methods may overlook by shifting the focus from raw price data to derived metrics. The specific targets of this study include predicting the next day's trend, identifying peaks and troughs, and estimating distances and heights related to these points. This innovative methodology represents a significant advancement over conventional forecasting techniques, allowing for a more nuanced understanding of market behavior. By extracting and analyzing a diverse array of derivative indices, this study aims to uncover hidden patterns within stock price time series that have not been previously explored. This innovative perspective could lead to the development of more robust predictive models, thereby contributing to the ongoing evolution of financial analytics and investment strategies. The proposed approach of this paper has several innovative aspects as follows:

- A wide variety of input variables has been considered in this study.
- Engineered and derivative indices for stock price prediction have extracted from daily stock price time series data. These indices are to be considered as target variables. These targets are:
 - o (1) the next day trend;
 - o (2) whether the next day will be a peak?
 - o (3) whether the next day will be a trough?
 - o (4) prediction of the distance between two consecutive peaks
 - o (5) prediction of the distance between two consecutive troughs
 - o (6) prediction of the distance from one peak or trough to the next.
 - o (7) prediction of the height of the next peak
 - o (8) prediction of the depth of the next trough
 - o (9) comparing the closing price of two consecutive trading day
 - o (10) the next day stock return
- To our best knowledge, the composition of these variables has not been used in previous researches. We are trying to extract a deep knowledge from the time series of stock price using these engineered indices. Combination of these indices will probably cover different hidden and complicated aspects of stock price time series based on a new perspective.
- It has been applied on a real case study including 5283 records of four stock prices from FTSE 100. So, the proposed model is a practical approach which can be used by many investors in order to automate short-term investment decision-making.
- In summary, this paper seeks to advance the understanding of stock price prediction through innovative methodologies and provides practical solutions that can significantly benefit investors in navigating the complexities of financial markets.

The paper is organized as follows. In Section 2, a literature review of past research works is presented. In Section 3, research methodology is presented. In Section 4, empirical results are represented. In Section 5, conclusions are presented.

2. Literature of past works

Successful prediction of price in the stock market has a significant impact on investors' profits. So, it has attracted the attention of many researchers. This section provides an overview of previous researches in the related fields.

Ayyildiz and Iskenderoglu (2024) conducted a study to evaluate the effectiveness of various machine learning algorithms in forecasting the directional movements of stock market indexes in developed countries. The research utilized a range of algorithms, including decision trees, random forests, k-nearest neighbors, naive Bayes, logistic regression, support vector machines, and artificial neural networks. The results indicated that artificial neural networks delivered the best average prediction performance among the algorithms tested, achieving accuracy rates exceeding 70% for several significant indices, including the NYSE 100 and FTSE 100. Ayyildiz (2023) explored the use of machine learning techniques to predict stock market index movements. It discusses various methodologies and models used in the prediction process, emphasizing the potential of machine learning to enhance forecasting accuracy. Ayyildiz and Iskenderoglu (2023) investigated the application of machine learning methods to predict stock index movement directions in developing countries. The authors analyzed various algorithms' effectiveness and applicability in different economic contexts. They also expanded the scope of machine learning applications in stock market predictions by focusing on developing economies, which may face various challenges and opportunities compared to developed markets. Ayyildiz and Iskenderoglu (2023) conducted a study focusing on predicting stock index movements using machine learning methods specifically applied to the BIST 100 Index, representing the top 100 companies listed on Borsa Istanbul. The study evaluates the performance of various machine learning models in this context.

Ma et al. (2021) applied random forest (RF), support vector regression (SVR), LSTM neural network, deep multilayer perceptron (DMLP) and convolutional neural network for the next day return prediction. Ribeiro, Santos, Mariani and Coelho (2021) proposed a novel hybrid model named HAR-PSO-ESN that combines heterogeneous autoregressive (HAR) specification, the echo state network (ESN), and the particle swarm optimization (PSO) metaheuristic for predicting realized volatility of stock prices. Kumar Chandar (2021) proposed nine new integrated models for forecasting intraday stock price based on the potential of three ANNs.

Paiva, Cardoso, Hanaoka and Duarte (2019) proposed a hybrid model using support vector machine (SVM) and mean-variance (MV) method for portfolio selection. SVM algorithm applied for predicting whether a stock has the potential to gain expected return (1% return in $t + 1$) or not. Zhang, Yan and Aasma (2020) proposed a hybrid deep learning model to predict the one-step-ahead closing price of stock indices. Vijh, Chandola, Tikkiwal and Kumar (2020) applied machine learning techniques such as artificial neural networks (ANN) and Random Forest (RF) for predicting the next day closing price of the stock. Yang et al. (2019) proposed a novel hybrid stock selection method. The authors applied extreme learning machine (ELM) in order to implement stock price prediction. Bisoi, Dash and Parida (2019) addressed stock price and trend prediction problems and using the

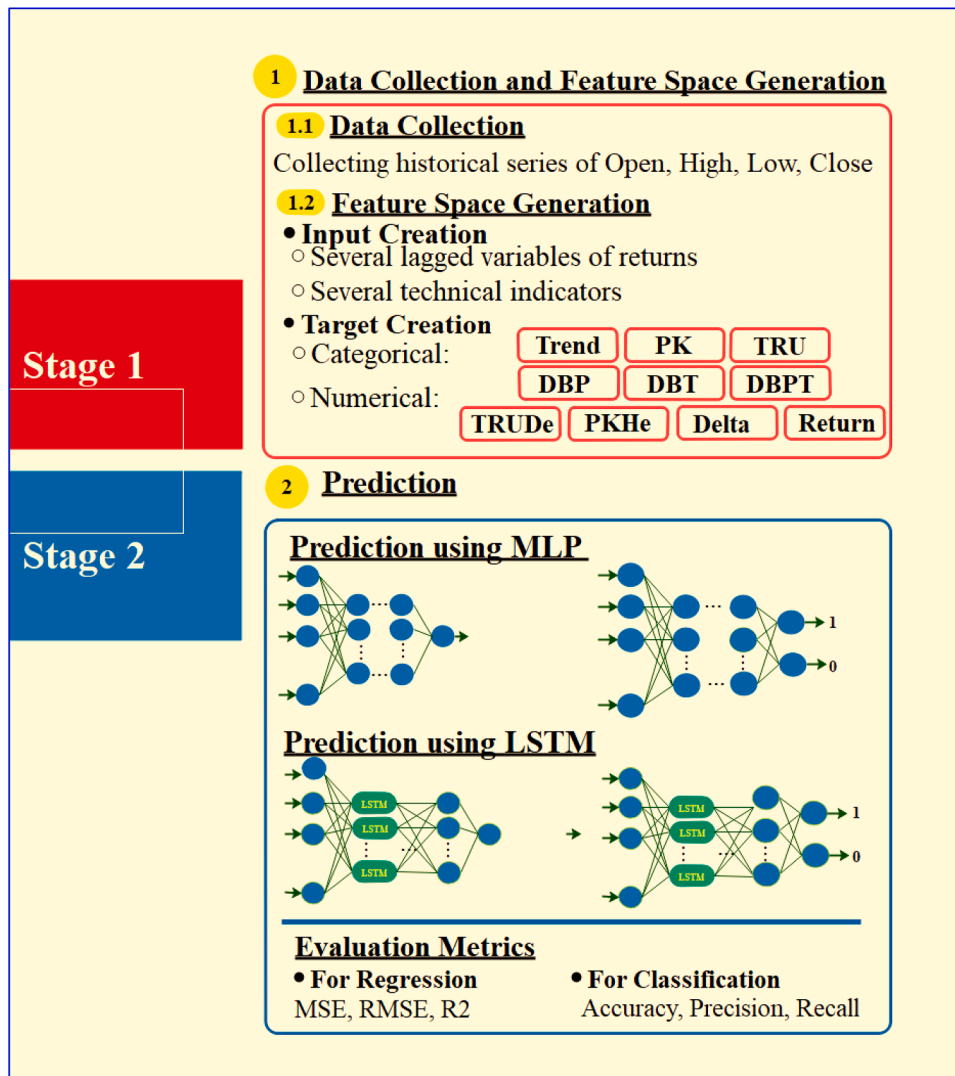


Fig. 1. Proposed Methodology of the Research.

combination of robust kernel extreme learning machine (RKELM) and variational mode decomposition (VMD) for this purpose. Ismail, Md Noorani, Ismail, Abdul Razak and Alias (2020) addressed the next day direction of stock price movement problem. The classification was implemented using logistic regression (LR), artificial neural network (ANN), support vector machine (SVM) and random forest (RF) algorithms. Tan et al. (2019) implement the RF algorithm in the field of stock selection strategy. Several fundamental/technical variables as well as momentum have been used to predict long-term and short-term stock price trend. Zhou, Zhou, Yang and Yang (2019) integrated Empirical Mode Decomposition (EMD) and Factorization Machine (FM) based Neural Network technique for stock market trend prediction. Basak et al. (2019) applied tree-based classifiers i.e. RF, and gradient boosted decision trees for predicting the direction of stock market prices.

Xu, Chhim, Zheng and Nojima (2020) proposed time series predictive model using integration of wavelet transform, the stacked auto-encoders and the bidirectional long-short term memory. Discrete wavelet transform was applied to de-noise the stock data, stacked auto-encoders were applied to extract deep features from technical indicators. The bidirectional long-short term memory (LSTM) applied as forecasting algorithm. A summary of previous studies in the field of stock prediction are listed in Table 1.

Statistical methods such as auto-regression moving average (ARMA) (Baek & Kim, 2018), autoregressive integrated moving average (ARIMA)

(Chen et al., 2021; Wang et al., 2020), Autoregressive conditional heteroscedasticity (ARCH) (Chen et al., 2021; Gandhmal & Kumar, 2019), generalized autoregressive conditional heteroskedasticity (GARCH) (Chen et al., 2021; Gandhmal & Kumar, 2019), and the smooth transition autoregressive model (STAR) (Zhou et al., 2019) have been previously applied to stock market prediction.

There are several studies that show the superiority of machine learning (ML) methods over classical methods in terms of financial time series predictions (Freitas, Souza & Almeida, 2009). Many researchers claimed that machine learning methods deal with the complex, nonlinear, multi-dimensional and noisy financial time series better than traditional statistical models (Baek & Kim, 2018; Chen et al., 2021; Wang et al., 2020). Different kind of ML methods such as support vector machines (SVM) (Chen & Hao, 2018; Lee, 2009; Paiva et al., 2019), support vector regression (SVR) (Lahmiri, 2018; Ma et al., 2021) neural networks (NNs) (Chen et al., 2021; Ismail et al., 2020; Zhou et al., 2019), Naive-Bayes (Patel, Shah, Thakkar & Kotecha, 2015), K-Nearest Neighbor (Ballings, Van Den Poel, Hespeels & Gryp, 2015; Subha, Nambi, 2012), ensemble learning i.e. Random forest (RF) (Ballings et al., 2015; Patel et al., 2015; Tan et al., 2019; Wang et al., 2020), AdaBoost (AB) (Ampomah, Qin, Nyame & Botchey, 2020; Ballings et al., 2015), Bagging (Ampomah et al., 2020) have been applied on stock market prediction.

In the last few years Deep Learning (DL) as a subfield of ML has been

introduced by many researchers as one of the best and most powerful methods of financial time series forecasting. Different type of deep learning algorithms including Deep Multi-Layer Perceptron (DMLP), Recurrent Neural Network (RNN) (Pawar, Jalem & Tiwari, 2019), Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNNs) have been widely applied on this subject (Sezer et al., 2020).

In order to effectively predict the future status of financial time series, in addition to choosing a strong forecasting method, selecting appropriate indicators and input attributes will play a significant role (Long, Lu & Cui, 2019; Wang et al., 2020). Most studies only used raw price data (Open, High, Low, and Close prices, and trading volume of assets) as the input to prediction models (Chen, Zhou & Dai, 2015; Dezsi & Nistor, 2016; Lee & Yoo, 2020). Many studies applied several lagged variables of return (Ma et al., 2021; Paiva et al., 2019; Wang et al., 2020). Some studies have used a variety of technical indicators as input (Dai, Zhu & Kang, 2021; Basak et al., 2019). Others have used fundamental variables (Abe & Nakayama, 2018). Some studies have used a combination of technical and fundamental indicators (Fu et al., 2018; Tan et al., 2019).

In this study, several lagged return variables and several technical indicators have been considered input variables.

Engineered and derivative indices for stock price prediction were extracted from daily stock price time series data and considered target variables. Engineered and derivative indices are specifically designed to capture intricate patterns in stock price movements that raw price data may not reveal. These indices can also encapsulate the complexities and non-linearity relationships in financial markets, enabling models to adapt more effectively to dynamic market conditions. Stock price data is often noisy due to market volatility and external factors. Engineered indices help filter this noise by emphasizing significant trends and relationships. By focusing on these indices, several specific investment questions can be addressed. For instance, predicting whether the next day will be a peak or trough provides actionable insights for traders looking to optimize their entry and exit points. Investors can better understand the rationale behind the model's outputs by providing insights into specific market behaviors, such as predicting the height of the next peak or the depth of the next trough. This interpretability is crucial for building trust and facilitating informed decision-making among investors.

By systematically incorporating a comprehensive set of engineered and derivative indices, this pioneering research aims to fill the gap in the literature regarding the use of advanced features in stock price prediction. This approach seeks to enhance the predictive power of models and provide a more nuanced understanding of market dynamics.

3. Proposed machine learning approach for stock price prediction

The structure of proposed machine learning approach is introduced in this section.

3.1. Proposed machine learning approach structure

As mentioned, the main purpose of this paper is to develop a machine learning approach to predict the next day's stock prices using artificial neural networks (ANNs) through a novel and comprehensive set of engineered and derivative indices. The proposed conceptual framework, as shown in Fig. 1, consists of two main stages.

In the first stage, we focus on data collection and the extraction and engineering of indices from the original stock price time series data. This process transforms raw data into meaningful features that enhance predictive accuracy. In the second stage, predictions are generated using Multi-Layer Perceptron (MLP) networks, leveraging these engineered indices to uncover deeper insights into market dynamics. In the following, the techniques and algorithms used in the proposed multi-stage machine learning approach for stock price prediction are briefly

described.

3.2. Conceptual structure of proposed machine learning approach

Fig. 1 presents the main stages, i.e., data collection, and feature space generation, and prediction for the machine learning approach for stock price prediction.

3.1.1. First stage: data collection and feature space generation

3.1.1.1. Data collection. Four well-known stocks (including: "BP p.l.c." (BP), "BHP Group Ltd." (BHP), "GlaxoSmithKline" (GSK), and "Vodafone Group Plc" (VOD) were selected from FTSE 100. Daily stock price of these stocks for the period from January 1, 2000 until December 31, 2020 were gathered from yahoo finance and considered as datasets of this research. The research period, was chosen due to its historical significance, encompassing critical market events such as the dot-com bubble and the 2008 financial crisis.

The abbreviations of the selected stocks are used for sake of brevity in the following sections. The initial data involves the daily historical series of open prices, close prices, the highest prices, the lowest prices, and the trading volume of assets. This timeframe provides a comprehensive dataset with 5283 records for each selected stock allowing for a robust analysis of stock price movements across varying economic conditions. After partitioning this dataset into training, validation, and test subsets, we trained our model to learn patterns in the data. The availability of consistent and reliable daily stock price data from reputable sources further supports the validity of the findings.

The FTSE 100 index is a capitalization-weighted index of the 100 most highly capitalized companies traded on the London Stock Exchange (LSE). By selecting stocks from this index, the research ensures that the chosen companies are among the most significant and most liquid in the UK market, making them representative of the overall market performance. The four chosen stocks, including "BP" representing the oil and gas sector, "BHP" from the mining sector, "GSK" in the pharmaceutical field, and "VOD" from telecommunications, provide a varied representation of the market. This variety enables the research to capture the distinct characteristics and dynamics of different sectors, thereby enhancing the generalizability of the findings across the market while more effectively considering the diverse factors that impact stock prices and market behavior. The selected stocks show varying volatility levels, enhancing market analysis. BP and BHP are highly sensitive to commodity price shifts, leading to greater volatility, while GSK and Vodafone maintain stability due to their strong market positions and consistent demand.

The selection of BP, BHP, GSK, and VOD for an investment portfolio is justified by their robust financial metrics and market positions, which offer a blend of stability and growth potential. BP and BHP, with their substantial market capitalizations of approximately \$90.32 billion and \$206 billion, respectively, are leaders in the oil and mining sectors, benefiting from global commodity demand and offering attractive dividend yields of 5.70% and 5.00%. GSK stands out in the pharmaceutical industry with a market cap of around \$36 billion and a solid dividend yield of 4.50%, driven by its significant investments in research and development, which position it well for future growth through innovation. Meanwhile, Vodafone, with a market cap of approximately \$34 billion and a dividend yield of 6.50%, provides exposure to the telecommunications sector, characterized by steady demand and technological advancements. Collectively, these stocks enhance portfolio diversification across different sectors and present opportunities for capital appreciation and income generation, making them compelling choices for investors seeking a balanced and resilient investment strategy.

3.1.1.2. Feature space generation. As mentioned, in order to make a

Table 2

Input variables used in this study.

Row	Attribute	Row	Attribute
1	$r1 = \ln \left(\frac{\text{Close Price}_i}{\text{Close Price}_{i-1}} \right)$	15	$r15 = \ln \left(\frac{\text{Low Price}_{i-3}}{\text{Open Price}_{i-3}} \right)$
2	$r2 = \ln \left(\frac{\text{Close Price}_{i-1}}{\text{Close Price}_{i-2}} \right)$	16	Relative Strength Index (RSI)
3	$r3 = \ln \left(\frac{\text{Close Price}_{i-2}}{\text{Close Price}_{i-3}} \right)$	17	Momentum
4	$r4 = \ln \left(\frac{\text{Close Price}_{i-3}}{\text{Close Price}_{i-4}} \right)$	18	True Range (TR)
5	$r5 = \ln \left(\frac{\text{High Price}_i}{\text{Open Price}_i} \right)$	19	Average True Range (ATR)
6	$r6 = \ln \left(\frac{\text{High Price}_i}{\text{Open Price}_{i-1}} \right)$	20	parabolic SAR
7	$r7 = \ln \left(\frac{\text{High Price}_i}{\text{Open Price}_{i-2}} \right)$	21	Commodity Channel Index (CCI)
8	$r8 = \ln \left(\frac{\text{High Price}_i}{\text{Open Price}_{i-3}} \right)$	22	Simple Moving Average (SMA)
9	$r9 = \ln \left(\frac{\text{High Price}_{i-1}}{\text{Open Price}_{i-1}} \right)$	23	Weighted Moving Average (WMA)
10	$r10 = \ln \left(\frac{\text{High Price}_{i-2}}{\text{Open Price}_{i-2}} \right)$	24	Hull Moving Average (HMA)
11	$r11 = \ln \left(\frac{\text{High Price}_{i-3}}{\text{Open Price}_{i-3}} \right)$	25	Average directional index (ADX)
12	$r12 = \ln \left(\frac{\text{Low Price}_i}{\text{Open Price}_i} \right)$	26	William's % R.
13	$r13 = \ln \left(\frac{\text{Low Price}_{i-1}}{\text{Open Price}_{i-1}} \right)$	27	% K
14	$r14 = \ln \left(\frac{\text{Low Price}_{i-2}}{\text{Open Price}_{i-2}} \right)$	28	% D-slow

prediction of stock price, a complete set of engineered and derivative indices are extracted from original stock price time series in this research. In this section the proposed engineered indices, their descriptions and the method of mapping original data set into the considered space are presented.

• Input Creation

The selection of input variables is one of the important issues in implementing stock price prediction approaches. We considered both lagged variables of return and technical indicators as inputs. The input variables used in this study are presented in Table 2 (Paiva et al., 2019; Wang et al., 2020).

The technical indicators used in this research and their calculation formula are presented in Table 3.

The use of logarithmic returns is grounded in financial theory, providing a normalized measure of price changes over time that stabilizes the variance of stock price changes, making the data more suitable for modeling. Logarithmic returns can be easily summed over time, allowing for more straightforward calculations of cumulative returns and facilitating interpretation in terms of percentage changes, which is intuitive for financial analysis. Technical indicators like RSI, Momentum, TR, ATR, SMA, HMA, ADX, William's %R, and Stochastic Oscillator provide insights into market sentiment, trend strength, volatility, and potential reversals, enabling the model to adapt to varying market conditions and identify trends, support/resistance levels, overbought or oversold conditions. The integration of various features enhances the understanding of market dynamics and significantly boosts the predictive power of the MLP model, contributing to financial forecasting knowledge.

The predictive features were meticulously designed to capture relevant market dynamics over appropriate time spans. Logarithmic returns were calculated using closing, high, low, and open prices over 1 to 4 days to reflect immediate price movements, crucial for short-term trading strategies while acknowledging that a longer historical context could yield additional insights. Technical indicators like ATR, TR, RSI, and ADX were computed over 14 days to assess market sentiment and

Table 3

Selected Technical indicators and their formula.

Technical Indicator	Attribute	References
Simple n-day moving average (here $n = 14$)	$\frac{C_t + C_{t-1} + \dots + C_{t-10}}{10}$	Boltona et al. (2015)
WMA (Weighted n-day moving average)	$\frac{((n) \times C_t + (n-1) \times C_{t-1} + \dots + C_n)}{(n + (n-1) + \dots + 1)}$	Kara, Acar Boyacioglu and Baykan (2011)
HMA (Hull Moving Average)	$WMA(2 * WMA(n/2) - WMA(n)).sqrt(n)$	Gorgulho, Neves and Horta (2011)
Momentum	$C_t - C_{t-n}$	Kara et al. (2011)
Stochastic %K	$\frac{C_t - LL_{t-n}}{HH_{t-n} - LL_{t-n}} \times 100$	Yao, Tan and POH (1999)
Stochastic %D	$\frac{\sum_{i=0}^{n-1} K_{t-i} \%}{n}$	Kara et al. (2011)
Slow %D	$\frac{\sum_{i=0}^{n-1} \%D_{t-i}}{n}$	Kara et al. (2011)
Larry William's R %	$\frac{H_n - C_t}{H_n - L_n} \times 100$	de Almeida, Reynoso-Meza and Arns Steiner (2016)
RSI (Relative Strength Index)	$100 - \frac{100}{1 + \left(\frac{\sum_{i=0}^{n-1} Up_{t-i}/n}{\sum_{i=0}^{n-1} Dw_{t-i}/n} \right)}$	Çaran-Moroşan (2011)
CCI (Commodity Channel Index)	$\frac{M_t - SM_t}{0.015 D_t}$	Verma, Verma and Prasad Sahu (2023)
TR (True Range)	$\max\{max_t, C_{t-1}\} - \min\{min_t, C_{t-1}\}$	McHugh et al., (2015)
ATR (Average True Range)	$\frac{1}{n} \sum_{i=1}^n TR_i$	McHugh et al., (2015)
Average Directional Index (ADX)	$100 \times \frac{EMA_t(n, abs(+DI - (-DI)))}{+DI + (-DI)}$	Bruni (2017)
Parabolic SAR	$SAR_{n+1} = SAR_n + AF(EP - SAR_n)$	Wilder (1978)

C_t is the closing price at time t , L_t is the low price at time t , H_t is the High price at time t , LL_t and HH_t mean lowest low and highest high in the last t , days t , Up_t means the upward price change, Dw_t means the downward price change at time

$$t.M_t : \frac{H_t + L_t + C_t}{3}, SM_t : \frac{(\sum_{i=1}^n M_{t-i+1})}{n}, D_t : \left(\sum_{i=1}^n |M_{t-i+1} - SM_t| \right).$$

$$up_t = max_t - max_{t-1}, dw_t = min_{t-1} - min_t.$$

$$\text{if } up_t > dw_t \text{ and } up_t > 0 \text{ then } +DM = up_t \text{ . otherwise } +DM = 0;$$

$$\text{if } dw_t > up_t \text{ and } dw_t > 0 \text{ then } -DM = dw_t \text{ . otherwise } -DM = 0;$$

$$+DI_t(n) = \frac{100 EMA_t(n, +DM)}{ATR_t(n)}, -DI_t(n) = \frac{100 EMA_t(n, -DM)}{ATR_t(n)}.$$

$EMA_t(n, X)$ denotes the exponential moving average of X over the last n periods. The AF value represents the acceleration factor. EP is the extreme point in a trend (highest point reached by a price during an uptrend or the lowest price reached during a downtrend).

trend strength, enhancing the model's adaptability to varying market conditions. Additionally, indicators such as CCI, SMA, WMA, and HMA were calculated over 20 days to capture broader trends and improve the model's robustness.

• Target Creation

In this section, a set of novel engineered indices is derived from daily stock price data to be tested as the target. Attributes entitled "Trend", "Peak Detection (PK)", "Trough Detection (TRU)", "Distance between two consecutive peaks (DBP)", "Distance between two consecutive Troughs (DBT)", "Distance from one peak or trough to the next (DBPT)", "Height of the next peak (PKHe)", and "Depth of the next trough (TRUDe)", "Delta", and "Daily Return (Ret.)" are developed based on original stock close price time series. The calculated attributes are then

Table 4

Set of Derivative indices that be tested as Target variables.

Row	Attribute	Description	Type
1	Trend	Stock price movement	Binomial (0,1)
2	PK	Peak detection	Binomial (0,1)
3	TRU	Trough detection	Binomial (0,1)
4	DBP	Distance between two consecutive peaks	Numeric
5	DBT	Distance between two consecutive troughs	Numeric
6	DBPT	Distance from one peak or trough to the next	Numeric
7	PKHe	Height of the next peak	Numeric
8	TRUDe	Depth of the next trough	Numeric
9	Delta	The difference between the closing price of the current period and the previous period	Numeric
10	Ret.	Daily return of the selected stocks	Numeric

used separately as target variable in order to explore the future pattern of price time series. These features are presented in Table 4.

The method of calculating each engineered index is described as follows. In all equations n refers to the number of records of sample dataset per each stock.

$$Trend_{t+1} = \begin{cases} 1 & \text{if } P_{t+1} > P_t \\ 0 & \text{otherwise} \end{cases} \quad t = 1, \dots, n \quad (1)$$

- **Trend:** This index indicates the direction of an asset's price. In this research, "Trend" is considered as a binomial variable. If P_t denotes the closing price of stock in period t , so that for each day if the closing price of the current period is greater than the closing price of the previous period, "Trend" it is equal to one, otherwise it is equal to zero.

- **Peak Detection:** Peak Detection is a binomial variable that indicates whether each point in the dataset is a peak or not. The peak is the point at which the uptrend has turned into downtrend. This index is abbreviated to PK for convenience in the following. PK is defined by Eq. (2).

$$PK_{t+1} = \begin{cases} 1 & \text{if } Trend_t = 1 \text{ and } Trend_{t+1} = 0, \quad t = 1, \dots, n \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

- **Trough Detection:** Trough Detection is a binomial variable that indicates whether each point in the dataset is a Trough or not. The

Trough is the point at which the downtrend has turned into uptrend. This index is abbreviated to TRU for convenience in the following. "TRU" is defined as Eq. (3).

$$TRU_{t+1} = \begin{cases} 1 & \text{if } Trend_t = 0 \text{ and } Trend_{t+1} = 1 \\ 0 & \text{otherwise} \end{cases} \quad t = 1, \dots, n \quad (3)$$

- **Distance Between Two Consecutive Peak:** This index indicates the distance between two consecutive peaks. If v denotes the number of peaks in the sample dataset and PK_i , $i = 1, \dots, v$ represents the i^{th} peak, then the distance between the $i + 1^{th}$ and i^{th} peaks is calculated using Eq. (4). This index is abbreviated to "DBP" for convenience in the following

$$DBP_k = t_{PK_{i+1}} - t_{PK_i} - 1 \quad k = 1, \dots, v - 1 \quad (4)$$

where t is denote the time period of occurrence of each peak.

- **Distance Between Two Consecutive Trough:** This index indicates the distance between two consecutive troughs. If w denotes the number of troughs in the sample dataset and TRU_j , $j = 1, \dots, w$ represents the j^{th} trough, then the distance between the $j + 1^{th}$ and j^{th} troughs is calculated using Eq. (5). This index is abbreviated to "DBT" for convenience in the following.

$$DBT_k = t_{TRU_{j+1}} - t_{TRU_j} - 1 \quad k = 1, \dots, w - 1 \quad (5)$$

where t is denote the time period of occurrence of each peak.

- **Distance from one peak or trough to the next:** This index indicates the distance from one peak to its next trough, or the distance from one trough to its next peak. If v and w denotes the number of peaks and troughs respectively, PK_i , $i = 1, \dots, v$ represents the i^{th} peak and TRU_j , $j = 1, \dots, w$ represents the j^{th} trough, then the distance between PK_i and TRU_j is calculated using Eq. (6). This index is abbreviated to "DBPT" for convenience in the following.

$$DBPT_k = \text{Distance between } PK_i \text{ and } TRU_j = f(x) = \begin{cases} t_{PK_i} - t_{TRU_j} - 1 & \text{if } t_{PK_i} > t_{TRU_j} \\ t_{TRU_j} - t_{PK_i} - 1 & \text{if } t_{PK_i} < t_{TRU_j} \end{cases} \quad (6)$$

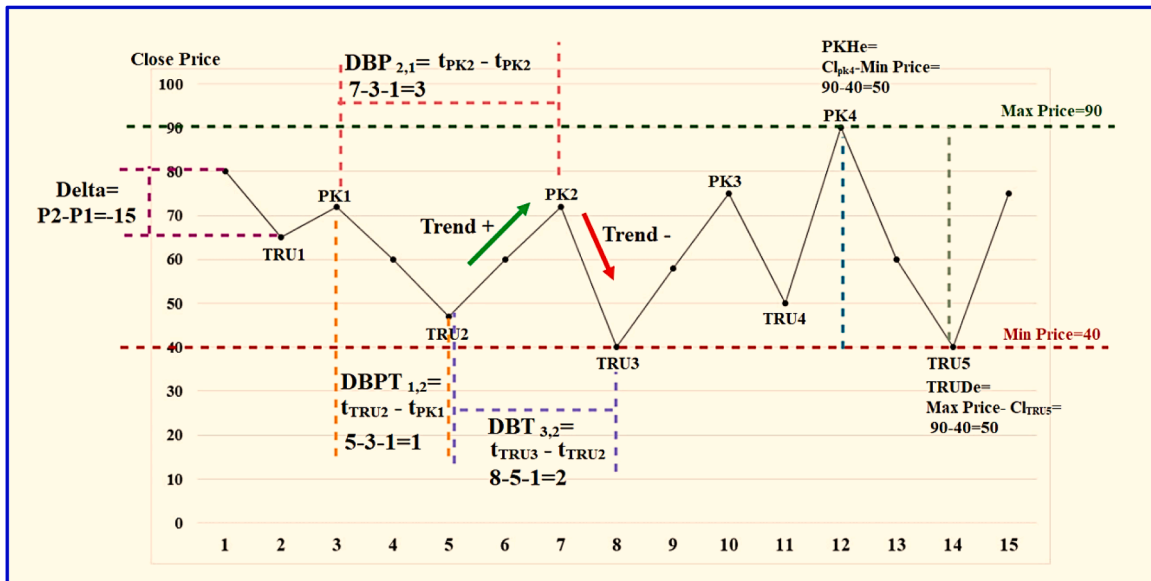


Fig. 2. Calculation method of the proposed indices from closing price.

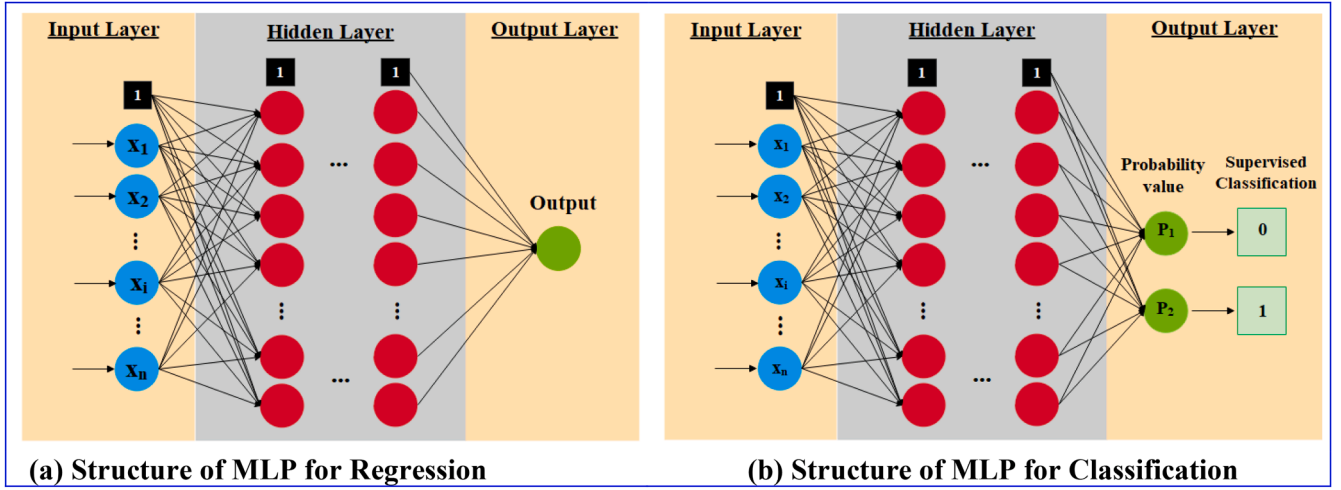


Fig. 3. ANNs' Structures.

where $k = 1. \dots u + v -$

- **Height of the Next Peak:** This index indicates the heights of a peak. If $\min CL_{all}$ denotes the minimum of the close price of sample dataset and CL_{PK_i} denotes the close price of the i^{th} peak, $i = 1. \dots v$, and $v < n$ in this case, the height of PK_i is calculated using Eq. (7). This index is abbreviated to PKHe for convenience in the following.

$$PKHe_i = CL_{PK_i} - \min CL_{all} \quad i = 1. \dots v, \quad v < n \quad (7)$$

- **Depth of the Next Trough:** This index indicates the Depth of a trough. If $\max CL_{all}$ denotes the maximum of the close price of sample dataset and CL_{TRU_j} denotes the close price of the j^{th} peak, $j = 1. \dots w$, $w < n$, in this case, the height of TRU_j is calculated using Eq. (8). This index is abbreviated to TRUDe for convenience in the following.

$$TRUDe_j = \max CL_{all} - CL_{TRU_j} \quad j = 1. \dots w, \quad w < n \quad (8)$$

- **Delta:** Delta represents the difference between the closing price of two sequential periods. Delta in this study is defined by Eq. (9).

$$\Delta_{t+1} = P_{t+1} - P_t \quad t = 1. \dots n \quad (9)$$

where P_t denotes the closing price of stock in period t .

- **Daily Return:** This index represents the stock daily return and measures the changes in a stock's price as a percentage of the previous day's closing price. This index is abbreviated to "Ret" for convenience in the following.

$$Ret_{t+1} = \left(\frac{P_{t+1} - P_t}{P_t} \right) \times 100 \quad t = 1. \dots n \quad (10)$$

Fig. 2 also represents the schematic view and calculation method of proposed indices.

3.1.2. Second stage: prediction

The derivative indices for all selected stocks for the next period are predicted thereby serving to reflect the future features of price time series and price prediction.

For ten derived indices including "Trend", "PK", "TRU", "DBP", "DBT", "DBPT", "PKHe", "TRUDe", "Delta", and "Ret." the prediction function for the next period is presented as follows.

$$\widehat{Target}_{i, t+1} = f(X_{1,t}, X_{2,t}, \dots, X_{28,t}), \quad i = 1.2 \dots 10 \quad (11)$$

Therefore, these indices are estimated based on its associated inputs

data $X_{k,t}$, $k = 1. \dots 28$. The $X_{k,t}$ represents the k^{th} element of the sample dataset for period t , $t = 1. \dots n$. The set of predictor variables was represented in Table 3.

3.1.2.1. Prediction using multi-layer perceptron. Artificial Neural Networks (ANN) is one of the most used methods in time series forecasting (Erzurum Cicek & Kamisli Ozturk, 2021). Multi-layer Perceptron (MLP) is a kind of artificial neural network which is relatively easy to implement. MLP has the ability to learn the complex relationship between the input variables and target due to the presence of activation function in each layer.

The Multi-Layer Perceptron (MLP) was selected as the prediction model for this study based on several critical criteria. First, MLPs are particularly effective at capturing the complex, non-linear relationships inherent in financial time series data, which is crucial for accurate stock price prediction. Additionally, as highlighted in Ayyıldız (2023), MLPs have demonstrated strong performance in predicting stock market movements, making them suitable for this research. The flexibility of MLPs to incorporate engineered and derivative indices as input features enhances their applicability, allowing for a more nuanced analysis of market dynamics. Furthermore, the study includes a comparative analysis with other machine learning models, such as logistic regression and support vector machines, to validate the effectiveness of the MLP and ensure a comprehensive evaluation of the predictive capabilities of different approaches. The computational cost associated with Multi-Layer Perceptron (MLP) architectures is typically lower than that of more complex machine learning models, such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks. This reduced computational burden enables faster training times and decreases resource requirements, making MLPs particularly well-suited for real-time applications in trading environments.

The following general objectives may be considered in multilayer perceptron training:

- **Regression.** In this case, the network regresses the independent variables, provided by the inputs, onto the dependent variable. The input variables can be categorical or numeric types, however, for regression ANNs, the dependent variable is numerical. The regression method will produce an estimated value of a dependent variable, which minimizes the mean-squared error.
- **Supervised classification.** In this case, the output variable is a categorical variable (or binary) and an input vector or pattern is to be trained to select a class from a list of classes.

The general topology of MLP consists of an input layer, one or more

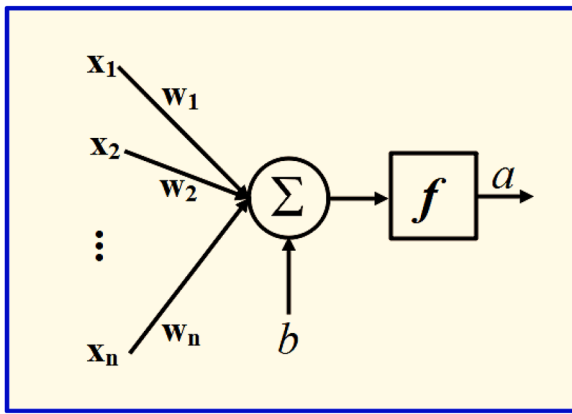


Fig. 4. Schematic view of a neuron.

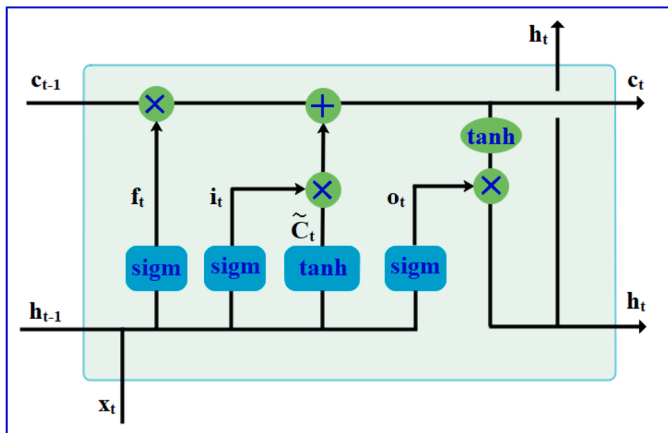


Fig. 5. LSTM structure.

Table 5
Confusion matrix.

		Predicted Class	
		P	N
Actual Class	P	TP	FN
	N	FP	TN

Table 6
The MLP model parameters setting.

Parameter	Classification	Prediction
Number of the input layer neurons	28	28
Number of neurons in hidden layer	10, 15, 20, 25, 30, 35, 40	10, 15, 20, 25, 30, 35, 40
Number of epochs	500, 1000	500, 1000
Performance Goal	1e-6, 1e-8, 1e-10, 1e-12	1e-6, 1e-8, 1e-10, 1e-12
Minimum Gradient	1e-06	1e-07
Maximum Validation Checks	6	20
Transfer function	tansig, softmax	tansig, purelin
Training function	Scaled Conjugate Gradient (SCG)	Levenberg–Marquardt training algorithm
Performance function	Cross-Entropy	MSE

Table 7
The LSTM model parameters setting.

Parameter	Value
Hidden layers	1,2,3,4
Hidden nodes	8, 32, 50, 64, 128, 256
Dropout rate	0.1–0.5
Activation Function	Relu, Sigmoid
Loss Function	mean_squared_error for regression, and binary_crossentropy for classification
Optimizer	adam
Batch Size	8, 10, 20, 30, 40, 50, 60, 70
Epochs	10, 15, 20, 25, 30, 40, 50, 128, 421, 524

Table 8
The results of trend classification using ANN model.

Stock	Dataset	Evaluation Metrics			
		Accuracy	Precision	Recall	F1 Score
BP	All Data	97.7	98.0	97.3	97.6
	Train	98.0	98.0	98.0	98.0
	Test	97.0	98.2	95.8	97.0
	Validation	97.0	98.2	95.7	96.9
BHP	All Data	96.9	96.7	96.9	96.8
	Train	97.0	96.9	97.0	96.9
	Test	95.8	95.2	95.9	95.5
	Validation	97.1	96.9	97.2	97.0
GSK	All Data	96.7	96.7	96.8	96.7
	Train	96.9	96.9	97.0	96.9
	Test	96.8	97.8	96.1	96.9
	Validation	95.8	95.0	96.2	95.6
VOD	All Data	96.1	96.4	96.0	96.2
	Train	96.4	96.6	96.4	96.5
	Test	96.6	96.7	96.5	96.6
	Validation	94.3	95.5	93.9	94.7

Table 9
The results of trend classification using LSTM.

Stock	Evaluation Metrics			
	Accuracy	Precision	Recall	F1 Score
BP	89	93	85	88.8
BHP	89	96	83	89.0
GSK	87	81	98	88.7
VOD	92	96	87	91.3

As can be seen, the performance of the ML model was better than the LSTM model.

Table 10
The results of peak detection using ANN model.

Stock	Dataset	Evaluation Metrics			
		Accuracy	Precision	Recall	F1 Score
BP	All Data	75.5	88.6	80.6	84.4
	Train	75.3	88.1	80.7	84.2
	Test	76.6	90.3	81.1	85.5
	Validation	75.3	89.3	80.0	84.4
BHP	All Data	75.6	93.0	78.4	85.1
	Train	75.4	93.0	78.1	84.9
	Test	75.9	93.9	78.2	85.3
	Validation	76.6	92.2	80.1	85.7
GSK	All Data	76.2	88.8	81.1	84.8
	Train	75.1	88.2	80.0	83.9
	Test	80.6	91.2	84.9	87.9
	Validation	76.6	88.8	82.2	85.4
VOD	All Data	75.3	88.8	80.0	84.2
	Train	75.9	88.5	81.1	84.6
	Test	73.6	89.3	77.2	82.8
	Validation	74.2	89.7	78.2	83.6

Table 11

The results of peak detection using LSTM.

Stock	Evaluation Metrics			
	Accuracy	Precision	Recall	F1 Score
BP	74	47	42	44.4
BHP	73	47	32	38.1
GSK	75	51	42	46.1
VOD	76	55	30	38.8

As can be seen, the performance of the MLP model based on the precision, recall, and F1 score indicators was better than that of the LSTM model.

Table 12

The results of trough detection using ANN model.

Stock	Dataset	Evaluation Metrics			
		Accuracy	Precision	Recall	F1 Score
BP	All Data	75.0	93.4	77.7	84.8
	Train	74.6	93.1	77.4	84.5
	Test	75.6	95.1	77.5	85.4
	Validation	76.4	93.3	79.7	86.0
BHP	All Data	76.1	87.3	81.9	84.5
	Train	76.7	88.0	82.1	84.9
	Test	75.1	85.8	82.2	84.0
	Validation	74.4	85.7	80.7	83.1
GSK	All Data	76.0	89.1	80.8	84.7
	Train	76.2	88.9	81.0	84.8
	Test	75.6	89.1	80.4	84.5
	Validation	75.6	89.9	80.2	84.8
VOD	All Data	74.3	87.0	80.0	83.4
	Train	74.6	87.2	80.3	83.6
	Test	74.2	87.3	79.5	83.2
	Validation	72.6	85.5	79.0	82.1

Table 13

The results of trough detection using LSTM.

Stock	Evaluation Metrics			
	Accuracy	Precision	Recall	F1 Score
BP	75	50	15	23.1
BHP	74	56	6	10.8
GSK	75	51	24	32.6
VOD	75	51	59	54.7

As can be seen, the performance of the MLP model based on the precision, recall and F1 score indicators was better than the LSTM model.

Table 14

The results of DBP prediction using ANN model.

Stock	Dataset	Evaluation Metrics		
		MSE	RMSE	R ²
BP	All	2.7741	1.6656	0.56291
	Train	2.1137	1.4539	0.65554
	Test	3.6385	1.9075	0.41247
	Validation	4.9816	2.232	0.35844
BHP	All	2.6967	1.6422	0.58192
	Train	2.3249	1.5248	0.656
	Test	3.2445	1.8013	0.4837
	Validation	3.8777	1.9692	0.34357
GSK	All	3.0307	1.7409	0.52903
	Train	2.3638	1.5375	0.65273
	Test	4.9127	2.2165	0.35139
	Validation	4.2529	2.0623	0.21827
VOD	All	2.3874	1.5451	0.55254
	Train	1.836	1.355	0.67592
	Test	2.8198	1.6792	0.4462
	Validation	4.5366	2.1299	0.22756

Table 15

The results of DBP prediction using LSTM.

Stock	Evaluation Metrics		
	MSE	RMSE	R ²
BP	0.0220	0.148	0.1350
BHP	0.0207	0.144	0.1759
GSK	0.0252	0.159	0.0006
VOD	0.0217	0.148	0.0361

As can be seen, LSTM based on the R² values is not capable of predicting this index.

hidden layers, and an output layer. In the input layer, a neuron is considered for each of the input variables. The hidden layer is used to show the nonlinear relationship between the variables and completes the data processing. Each hidden layer can have any number of neurons. The output layer is producing the final result. The output layer is aimed to produce the categorical class label or is used to predict continuous target indices. In Regression cases, the output layer will have only one neuron. In classification cases, the output layer should have neurons exactly the same number of classes of the target variable. (Huang, Lai, Nakamori & Wang, 2004; Murtagh, 1991; Siswanto, Prabuwo, Abdullah & Idrus, 2016)

The schematic architecture of MLP for regression and classification problems is presented in Fig. 3.

The structure of a neuron consists of inputs $x_1, x_2, \dots, x_n$, weights $w_1, w_2, \dots, w_n$, bias b , transfer function f , and output a is presented in Fig. 4.

The neuron inputs come from the other neurons in the previous layer. All weighted inputs and biases are summed and the result is passed through the transfer function to generate output, as in Eqs. (1) and (2). The output is sent to other neurons in the next layer as input.

$$S = \sum_{i=1}^n w_i x_i + b \quad (12)$$

$$a = f(s) \quad (13)$$

A hidden layer receives input from the previous layer and provides output to another hidden layer or an output layer. Various nonlinear activation functions such as Rectified Linear Activation (ReLU), Logistic (Sigmoid), Hyperbolic Tangent (Tanh) can be used in the hidden layers of a neural network. The activation function in the output layer must be selected according to the type of target variable. In a regression problem, a linear activation function should be used. In classification problems Soft-max, and sigmoid activation functions can be used. If the activation function is Sigmoid or Soft-max, the probability value is received from all nodes of the output layer. For the training process of ANNs, several learning techniques such as Levenberg-Marquardt (LM) and Scaled Conjugate Gradient (SCG) can be used for regression and classification problems respectively. The Mean Squared Error, or MSE, is the loss that can be used for regression problems and Cross-entropy is the loss function that can be used for classification problems.

3.1.2.2. Prediction using long short term memory (LSTM) network. To Justify and compare the effectiveness of MLP in forecasting the engineered indices, the Long Short-term memory (LSTM) network is applied to benchmark the proposed MLP model of the research.

LSTM is a recurrent neural network (RNN) that shows long-term dependencies in time series data. LSTM cell contains extra gates, namely the input, forget, and output gates. Estimation at the t stage is known based on the type of the information gate, forget gate, output gate, and cell gate (Wang et al., 2020).

Fig. 5 shows the structure of an LSTM memory cell.

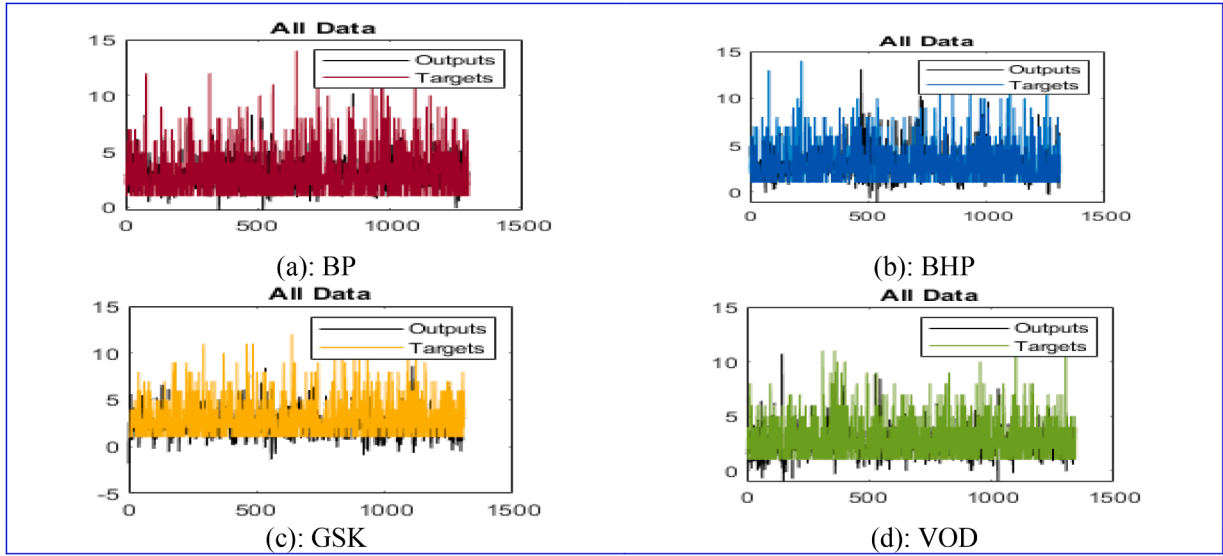


Fig. 6. Comparing the actual and the predictions of the ANN model for DBP.

Table 16

The results of DBT Prediction using ANN model.

Stock	Dataset	Evaluation Metrics		
		MSE	RMSE	R ²
BP	All	2.9245	1.7101	0.51853
	Train	2.3042	1.5179	0.62628
	Test	3.6571	1.9124	0.3878
	Validation	5.077	2.2532	0.31447
BHP	All	2.7854	1.669	0.53692
	Train	2.7102	1.6463	0.57008
	Test	3.1475	1.7741	0.53562
	Validation	2.773	1.6652	0.42146
GSK	All	2.9393	1.7144	0.45242
	Train	2.6199	1.6186	0.4939
	Test	4.5269	2.1277	0.29305
	Validation	2.8401	1.6853	0.46125
VOD	All	2.2899	1.5132	0.54817
	Train	2.1655	1.4716	0.60505
	Test	2.5986	1.612	0.42733
	Validation	2.5599	1.6	0.3829

Table 17

The results of DBT Prediction using LSTM.

Stock	Evaluation Metrics		
	MSE	RMSE	R ²
BP	0.0180	0.134	0.1157
BHP	0.0306	0.175	0.1097
GSK	0.0183	0.136	0.0140
VOD	0.0171	0.131	0.0391

As can be seen, LSTM based on the R² values is not capable of predicting this index.

In Fig. 5 *sigm* and *tanh* respectively denote a sigmoid and hyperbolic tangent activation functions.

The equations that describe the behavior of all components in the LSTM cell are described in the following.

The input of the information gate (at stage *t*) is calculated by Eq. (14).

$$i_t = \sigma(\mathbf{W}_{ih} \cdot \mathbf{h}_{t-1} + \mathbf{W}_{ix} \cdot \mathbf{x}_t) \quad (14)$$

The non-linear σ function is similar to a sigmoid function. $\mathbf{W}_{ih}$ and $\mathbf{W}_{ix}$ are weigh matrices that connect previous hidden layer to current

hidden layer and inputs to the hidden layer respectively.

Similarly, the input of forget gate and the input of output gate are calculated by the Eq. (15) and Eq. (16), respectively.

$$f_t = \sigma(\mathbf{W}_{fh} \cdot \mathbf{h}_{t-1} + \mathbf{W}_{fx} \cdot \mathbf{x}_t) \quad (15)$$

$$o_t = \sigma(\mathbf{W}_{oh} \cdot \mathbf{h}_{t-1} + \mathbf{W}_{ox} \cdot \mathbf{x}_t) \quad (16)$$

where $\mathbf{W}_{fh}$, $\mathbf{W}_{fx}$, $\mathbf{W}_{oh}$, $\mathbf{W}_{ox}$ are weight matrices.

The internal memory of the unit is calculated by Eq. (17).

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (17)$$

where C_{t-1} represents the cell status from the previous stage and $\tilde{C}_t$ is a candidate hidden state that is computed by Eq. (18).

$$\tilde{C}_t = \tanh(\mathbf{W}_{ch} \cdot \mathbf{h}_{t-1} + \mathbf{W}_{cx} \cdot \mathbf{x}_t) \quad (18)$$

Finally, h_t which is correspond to hidden state is calculated as using Eq. (19).

$$h_t = O_t \cdot \tanh(C_t) \quad (19)$$

• Evaluation Metrics

Evaluation metrics for prediction models: To evaluate the performance of continuous prediction models, three metrics including mean squared error (MSE), root mean squared error (RMSE), and R squared (R²) are applied. These metrics are defined as follows.

Mean square error (MSE): It is an indicator measuring the average squared difference between the observed values and predicted values (Ma et al., 2021; Vijn et al., 2020).

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (20)$$

Root-mean-square error (RMSE): It is another effective indicator measuring differences between the observed values and predicted values.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (21)$$

Coefficient of determination (R²): This is a measure of how well the model can be explained.

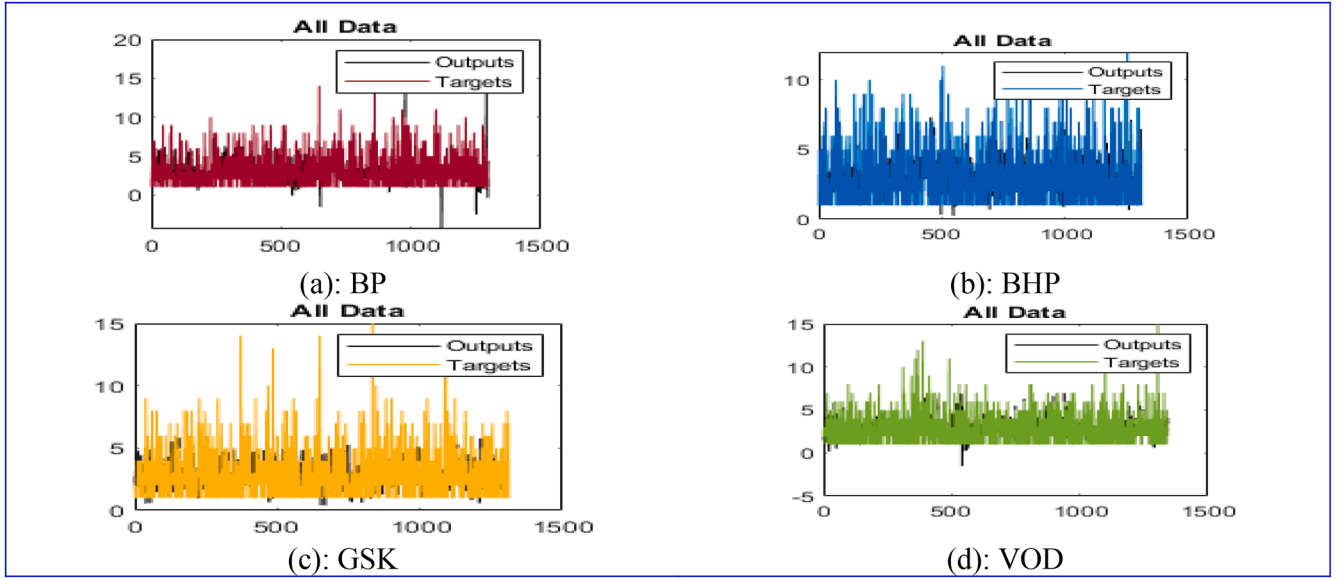


Fig. 7. Comparing the actual and the predictions of the ANN model for DBT.

Table 18
The results of DBPT Prediction using ANN model.

Stock	Dataset	Evaluation Metrics		
		MSE	RMSE	R ²
BP	All	0.7905	0.8891	0.77848
	Train	0.73511	0.85739	0.80403
	Test	0.92596	0.96227	0.71201
	Validation	0.91368	0.95587	0.70632
BHP	All	0.74249	0.86168	0.7878
	Train	0.69034	0.83087	0.8003
	Test	1.044	1.0218	0.70591
	Validation	0.68441	0.82729	0.82143
GSK	All	0.8889	0.94282	0.7276
	Train	0.83886	0.91589	0.75708
	Test	0.9195	0.95891	0.62738
	Validation	1.0914	1.0447	0.67623
VOD	All	0.77027	0.87765	0.73033
	Train	0.59065	0.76854	0.79227
	Test	1.1288	1.0624	0.61827
	Validation	1.2502	1.1181	0.57893

Table 19
The results of DBPT Prediction using LSTM.

Stock	Evaluation Metrics		
	MSE	RMSE	R ²
BP	0.00979	0.099	0.2295
BHP	0.0108	0.104	0.500
GSK	0.0177	0.133	0.1193
VOD	0.008	0.090	0.3454

As can be seen, LSTM based on the R² values is not capable of predicting this index.

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (22)$$

Evaluation metrics for classification models: In order to evaluate classification models three metrics i.e. as Accuracy, Precision, Recall, and F1-score are used. These metrics are calculated using True Positive (TP), False Positive (FP), True Negative (TN) and False Negative (FN). The formula of these metric is depicted in Eqs. (23) – (26) (Luque, Carrasco, Martína & de las Heras, 2019) (Table 5).

$$Accuracy = \frac{TN + TP}{TP + TN + FP + FN} \quad (23)$$

$$Precision = \frac{TP}{TP + FP} \quad (24)$$

$$Recall = \frac{TP}{TP + FN} \quad (25)$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (26)$$

4. Empirical results

The results of proposed MLP for classification and prediction of designed engineered and derivative indices have been presented in this section.

4.1. Structure of proposed MLP

The proposed MLP is implemented in Matlab R2018b in this study. Two distinct Multi-layer perceptron have been used to perform prediction of continuous indices and classification of binary variables. “Trend”, “PK”, and “TRU” are binary target indices and “Delta”, “Return”, “PKHe”, “TRUDe”, “DBP”, “DBT”, and “DPBT” are continuous target indices. Each MLP was employed for each target and all four selected stocks utilizing a comprehensive dataset of 5283 records for each stock. These records are related to the daily historical time series of stock prices between January 1, 2000, and December 31, 2020. All variables which were presented in Table 2 are assumed as input of both MLPs. The comprehensive dataset of 5283 records for each selected stock was divided into three distinct subsets: 70% for training, 15% for validation, and 15% for testing. The test dataset, comprising approximately 792 records for each stock, is reserved for evaluating the model’s performance on unseen data, ensuring a robust assessment of its predictive capabilities. Meanwhile, the validation dataset consists of about 792 records and is utilized during the training process to fine-tune model hyperparameters. This structured approach to dataset partitioning allows our research to enhance predictive capabilities through engineered and derivative indices while providing valuable insights into stock price dynamics. This division is the most well-known setting for portioning the data patterns.

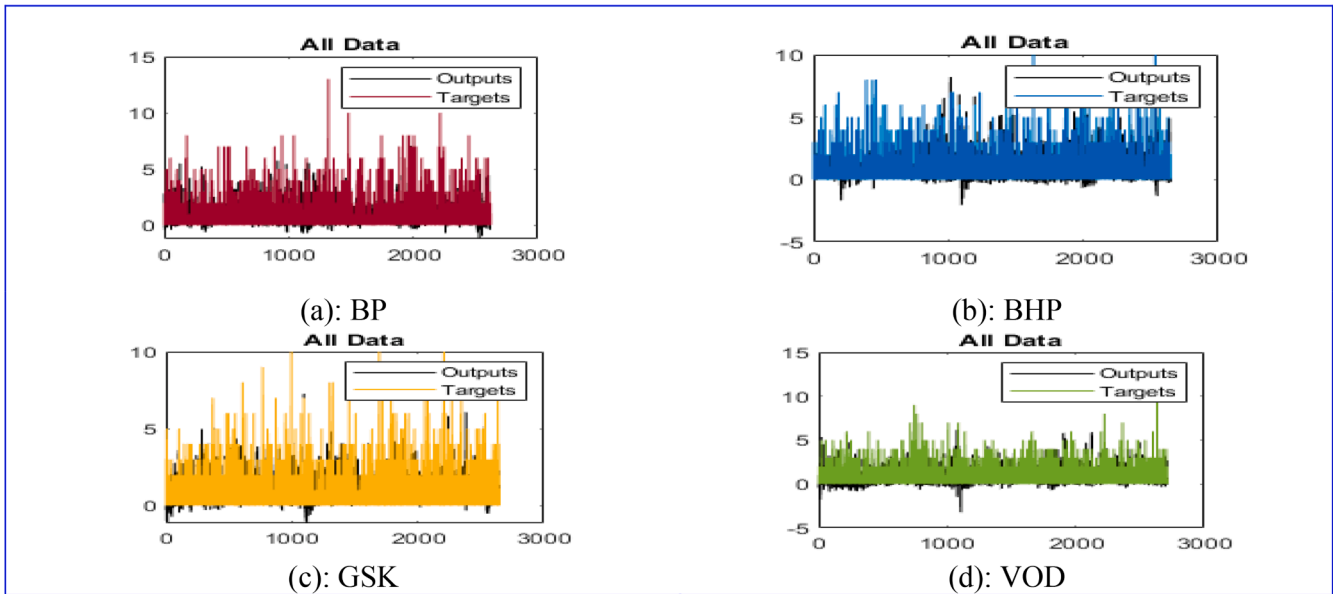


Fig. 8. Comparing the actual and the predictions of the ANN model for DBPT.

Table 20

The results of prediction of peak height using ANN model.

Stock	Dataset	Evaluation Metrics		
		MSE	RMSE	R ²
BP	All	0.024125	0.15532	0.99992
	Train	0.018731	0.13686	0.99994
	Test	0.047378	0.21766	0.99982
	Validation	0.026013	0.16129	0.99992
BHP	All	0.098238	0.31343	0.99992
	Train	0.10059	0.31716	0.99992
	Test	0.083997	0.28982	0.99993
	Validation	0.10153	0.31864	0.99991
GSK	All	0.021662	0.14718	0.99976
	Train	0.010157	0.10078	0.99999
	Test	0.055278	0.23511	0.99942
	Validation	0.041544	0.20382	0.99952
VOD	All	0.016971	0.13027	0.99987
	Train	0.0038197	0.061804	0.99997
	Test	0.042791	0.20686	0.99962
	Validation	0.052353	0.22881	0.99969

Table 21

The results of prediction of peak height using LSTM.

Stock	Evaluation Metrics		
	MSE	RMSE	R ²
BP	0.0036	0.060	0.7800
BHP	0.0004	0.022	0.902
GSK	0.0003	0.018	0.93743
VOD	0.0009	0.031	0.8951

As can be seen, the performance of the MLP model was better than the LSTM model.

The Taguchi method is implemented in this study to optimize the parameters of the Multi-Layer Perceptron (MLP) neural network. The key parameters to be optimized are the number of neurons in the hidden layer, learning rate, performance goal, and maximum validation failures. The Taguchi method involves systematically designing an orthogonal array to explore the parameter space. Seven levels of hidden layer sizes (10, 15, 20, 25, 30, 35, and 40) and four levels each for learning rates (0.01, 0.05, 0.1, 0.2), and four levels for performance

goals (1e-6, 1e-8, 1e-10, 1e-12), and maximum validation failures (5, 10, 20, 30) was considered. For each experiment, the MLP network is trained with the specified parameter settings, and the performance is evaluated using the Mean Squared Error (MSE) as the performance metric. The signal-to-noise ratio (SNR) is calculated for each experiment to maximize the SNR and achieve optimal performance. The code analyzes the results of the experiments by calculating the mean SNR for each parameter at each level. The combination of optimal levels for all parameters is determined as the overall optimal parameter setting.

For categorical indices Tangent sigmoid (Tansig) was used as an activation function at hidden layer nodes and 'Soft-max' was used as a linear transfer function at the output layer.

Tangent sigmoid (Tansig) was used as an activation function at hidden layer nodes and 'Purelin' was used as a linear transfer function at the output layer for continuous target indices.

Levenberg–Marquardt (LM) method was used as training algorithm for prediction models and Scaled Conjugate Gradient (SCG) was used as training algorithm for prediction models. The parameters of the both MLPs are briefly shown in Table 6.

The results of ten target derivative indices are presented in the following. Modeling was performed for all 4 selected stocks separately.

First, the results of classification of binomial indices including 'Trend', 'PK', and 'TRU' are presented.

4.2. Structure of LSTM network

In this research, the LSTM method has been used to perform the prediction step. The LSTM model is coded in Python 3.8.6.

The studying period is divided into a training subperiod (the first 70% of the whole period) and a testing subperiod (the remaining 30%). The training examples are normalized element-wise using min-max scaling.

A couple of modules from Keras are applied to build the LSTM model. Sequential is used to initialize the neural network, and LSTM is used to add the Long Short-Term Memory layer. Dropout for adding dropout layers that prevent overfitting. Dense for adding a densely connected neural network layer.

LSTM parameters such as hidden layers, nodes, dropout rate, number of epochs, and batch size were investigated for different values. For LSTM layers, the "Relu" activation function and the "Sigmoid" activation function are applied for Dense layers. "Mean_squared_error" is

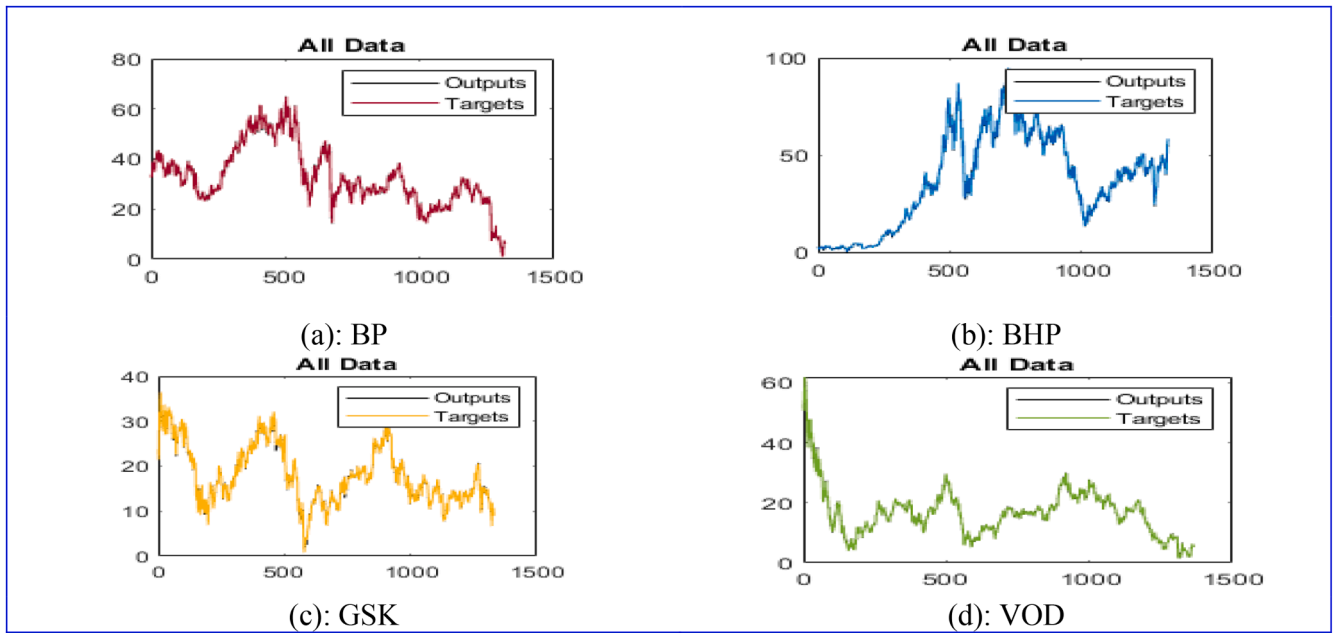


Fig. 9. Comparing the actual and the predictions of the ANN model for PKHe.

Table 22
The results of prediction of trough depth using ANN model.

Stock	Dataset	Evaluation Metrics		
		MSE	RMSE	R ²
BP	All	0.043989	0.20973	0.99984
	Train	0.043643	0.20891	0.99984
	Test	0.05226	0.22861	0.99981
	Validation	0.037325	0.1932	0.99988
BHP	All	0.081624	0.2857	0.99993
	Train	0.070385	0.2653	0.99994
	Test	0.11383	0.33739	0.9999
	Validation	0.10173	0.31896	0.9999
GSK	All	0.028403	0.16853	0.99965
	Train	0.027001	0.16432	0.99967
	Test	0.032709	0.18086	0.99962
	Validation	0.030614	0.17497	0.99963
VOD	All	0.030149	0.17363	0.99975
	Train	0.028467	0.16872	0.99977
	Test	0.032544	0.1804	0.9997
	Validation	0.035579	0.18862	0.99971

Table 23
The results of prediction of trough depth using LSTM.

Stock	Evaluation Metrics		
	MSE	RMSE	R ²
BP	0.0027	0.053	0.8444
BHP	0.0002	0.015	0.95550
GSK	0.0003	0.020	0.9307
VOD	0.0002	0.017	0.9721

As can be seen, the performance of the MLP model was better than the LSTM model.

considered a loss function of regression problems, and “binary_crossentropy” is regarded as a loss function of classification problems. Adam optimizer is used to update the weights of networks. The parameters related to the structure of LSTM and its implementation are shown in Table 7.

Two distinct LSTM have been used to perform prediction of continuous indices and classification of binary variables. Each LSTM was

employed for each target and all four selected stocks. The results of the modeling with respect to the ten derivative indices are presented in the following.

4.3. Trend classification

4.3.1. Trend classification using ANN model

The results of closing price trend classification using MLP are presented in Table 8. As trend is a binary target, the performance of the classification MLP is evaluated using Accuracy, Precision, and Recall. As can be seen, the trend forecasting has been done with high accuracy.

4.3.2. Trend classification using LSTM

The results of closing price trend classification using LSTM are presented in Table 9.

4.4. Peak detection (PK)

4.4.1. Peak detection (PK) using ANN model

As mentioned, the “PK” variable indicates whether the closing price will be at the peak point the next day or not. In fact, PK is a binary variable, where PK=1 if the next day is peak and PK= 0 otherwise. The results are shown in Table 10. As can be seen the value accuracy, precision and recall metrics for all selected stocks are suitable.

4.4.2. Peak detection (PK) using LSTM

The results of Peak Detection (PK) using LSTM are presented in Table 11.

4.5. Trough detection

4.5.1. Trough detection using ANN model

As mentioned, the “TRU” variable indicates whether the closing price will be at the trough point the next day or not. In fact, TRU is a binary variable, where if TRU=1 the next day is trough and if TRU = 0 otherwise. The performance of the proposed ANN is represented in Table 12.

4.5.2. Trough detection using LSTM

The results of Trough Detection using LSTM are presented in Table 13.

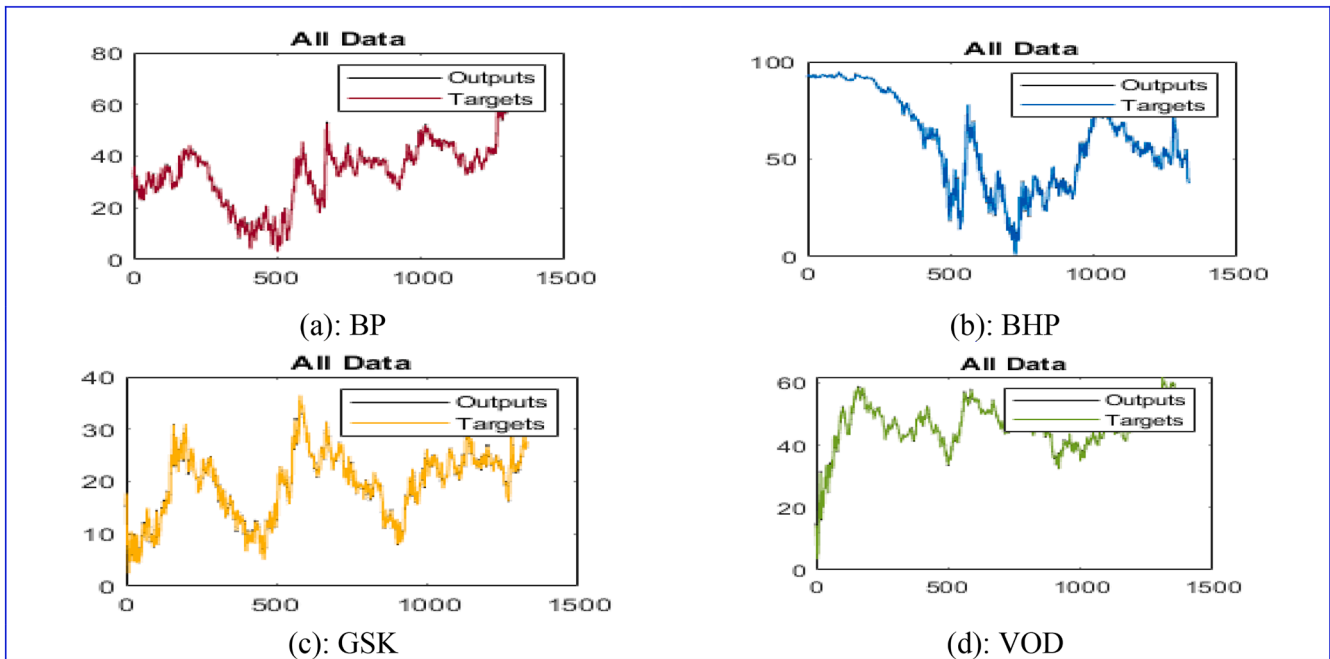


Fig. 10. Comparing the actual and the predictions of the ANN model for TRUDE.

Table 24
The results of Delta prediction using ANN model.

Stock	Dataset	Evaluation Metrics		
		MSE	RMSE	R ²
BP	All Data	0.0001664	0.0129	0.99988
	Train	0.00012249	0.011067	0.99991
	Test	0.00036693	0.019155	0.99972
	Validation	0.00017089	0.013073	0.99989
BHP	All Data	7.3115e-05	0.0085507	0.99997
	Train	5.948e-05	0.0077123	0.99998
	Test	0.00011297	0.0106297	0.99996
	Validation	9.6921e-05	0.0098448	0.99996
GSK	All Data	1.1134e-05	0.0033367	0.99999
	Train	4.77e-06	0.002184	0.99999
	Test	1.8325e-05	0.0042807	0.99998
	Validation	3.3653e-05	0.0058011	0.99995
VOD	All Data	3.256e-05	0.0057061	0.99995
	Train	1.3412e-05	0.0036623	0.99989
	Test	0.00010458	0.010226	0.99998
	Validation	4.9939e-05	0.0070668	0.99993

Table 25
The results of Delta prediction using LSTM.

Stock	Evaluation Metrics		
	MSE	RMSE	R ²
BP	0.00016	0.013	0.9230
BHP	0.00015	0.013	0.95042
GSK	0.00012	0.011	0.9585
VOD	2.00e-05	0.004	0.9636

As can be seen, the performance of the MLP model was better than the LSTM model.

In the following, the results of prediction of numerical indices including 'DBP', 'DBT', 'DBPT', 'PKHe', 'TRUDE' 'Delta', and 'Ret.', are presented.

4.6. DBP prediction results

4.6.1. DBP prediction results using ANN model

The results of predicting the distance between two consecutive peaks (DBP) using MLP is presented in Table 14. The R² values are greater than 0.22 for all stocks in train, test, and validation data.

4.6.2. DBP prediction results using LSTM

The results of DBP Prediction using LSTM are presented in Table 15. Fig. 6 presents the real target and the output of ANN for "DBP" in four selected stocks. The differences between target and output can be detected.

4.7. DBT prediction results

4.7.1. DBT prediction results using ANN model

The results of predicting the distance between two consecutive troughs (DBT) using MLP model are presented in Table 16.

4.7.2. DBT prediction results using LSTM

The results of DBP Prediction using LSTM are presented in Table 17.

Fig. 7 comparing the real target and the output of ANN model for "DBT" in four selected stocks.

4.8. DBPT prediction results

4.8.1. DBPT prediction results using ANN model

The results of predicting the distance between two consecutive peak and troughs (DBPT) using MLP model are presented in Table 18.

4.8.2. DBPT prediction results using LSTM

The results of predicting the distance between two consecutive peak and troughs (DBPT) using LSTM are presented in Table 19.

Fig. 8 presents the real target and the output of ANN for "DBPT" in four selected stocks.

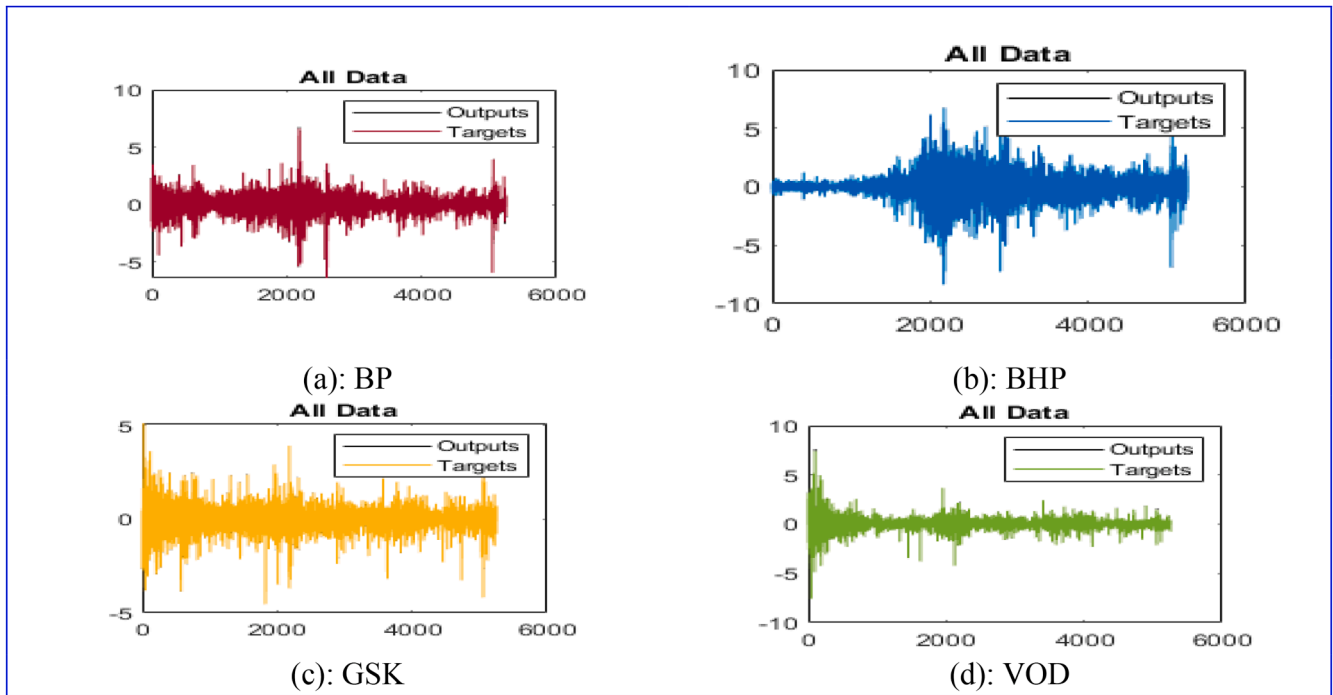


Fig. 11. Comparing the actual and the predictions of the ANN model for Delta.

Table 26

The results of return prediction using ANN model.

Stock	Dataset	Evaluation Metrics		
		MSE	RMSE	R ²
BP	All Data	5.9796e-08	0.00024453	0.99992
	Train	6.2174e-09	7.885e-05	0.99999
	Test	3.1725e-07	0.00056325	0.99973
	Validation	5.249e-08	0.00022911	0.99993
BHP	All Data	1.3216e-08	0.0001496	0.99999
	Train	8.5859e-09	9.266e-05	0.99999
	Test	1.5154e-08	0.0001231	0.99999
	Validation	3.8754e-09	6.2253e-05	0.99999
GSK	All Data	7.5035e-08	0.00027393	0.99982
	Train	8.7082e-09	9.9318e-05	0.99998
	Test	1.9731e-07	0.00044419	0.99952
	Validation	2.6243e-07	0.00051228	0.99928
VOD	All Data	1.0908e-08	0.00010444	0.99999
	Train	9.8742e-09	9.9369e-05	0.99999
	Test	1.0343e-08	0.0001017	0.99999
	Validation	1.6297e-08	0.00012766	0.99998

Table 27

The results of return prediction using LSTM.

Stock	Evaluation Metrics		
	MSE	RMSE	R ²
BP	0.0002	0.016	0.909
BHP	0.0001	0.011	0.962
GSK	0.0001	0.013	0.9515
VOD	0.0001	0.011	0.9674

As can be seen, the performance of the MLP model was better than the LSTM model.

4.9. Peak height prediction

4.9.1. Peak height prediction using ANN model

The result of predicting the height of the next peak (PKHe) using MLP model is presented in Table 20. The R² values are greater than 0.9995 for

all stocks in train, test, and validation data.

4.9.2. Peak height prediction using LSTM

The result of predicting the height of the next peak (PKHe) using LSTM is presented in Table 21.

Fig. 9 presents the real target and the output of ANN model for “PKHe” in four selected stocks.

4.10. Trough depth prediction

4.10.1. Trough depth prediction using ANN model

The results of predicting the depth of the next trough (TRUDe) using MLP model are presented in Table 22. The R² values are greater than 0.9996 for all stocks in train, test, and validation data.

4.10.2. Trough depth prediction using LSTM

The results of predicting the depth of the next trough (TRUDe) using MLP model are presented in Table 23.

Fig. 10 presents the real target and the output of ANN model for “TRUDe” in four selected stocks.

4.11. Delta prediction

4.11.1. Delta prediction using ANN model

Delta compares the closing stock price of day $i + 1$ with the closing price of the previous trading day. The prediction performance of the proposed MLP models in terms of MSE, RMSE, and R², implemented on the 4 selected stocks are shown in Table 24. As can be seen, the prediction model has a good performance. The R² values are greater than 0.9998 for all stocks in train, test, and validation data.

4.11.2. Delta prediction using LSTM

The results of Delta Prediction using LSTM are presented in Table 25.

Fig. 11 comparing the real target and the output of proposed ANN for “Delta” in four selected stocks.

As clear the output and target values are exactly the same for more than 99.998 percentage of data patterns.

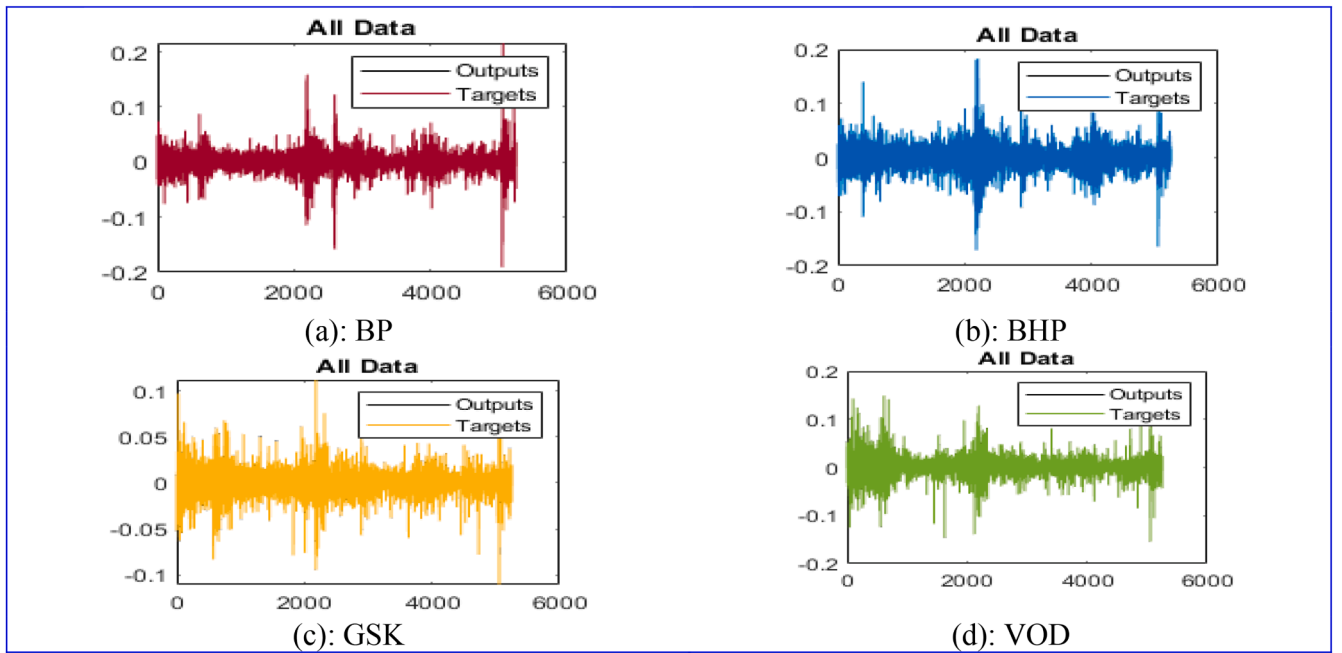


Fig. 12. Comparing the actual and the predictions of the ANN model for Return.

Table 28

The predicted values of the proposed indices for the new 19 days for BHP stock.

Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
	Original Closing Price (CP1)	Original Trend	Predicted Trend	Original PK	Predicted PK	Original TRU	Predicted TRU	Original DBPT	Predicted DBPT	Original PKHe	Predicted PKHe	Original TRUDE	Predicted TRUDE	Original Delta	Predicted Delta	Original Return	Predicted Return	Calculated Closing Price using Predicted Delta (CP2)	Calculated Closing Price using Predicted Return (CP3)
Dec 31, 2020	65.34	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
t+1	67.35	1	1	0	0	0	0							2.01	2.017	0.031	0.031	67.357	67.366
t+2	69.5	1	1	0	0	0	0							2.150	2.155	0.0319	0.032	69.512	69.521
t+3	71.71	1	1	0	0	0	0							2.201	2.200	0.0318	0.032	71.712	71.746
t+4	73.18	1	1	0	0	0	0							1.470	1.471	0.0205	0.020	73.183	73.181
t+5	73.46	1	1	1	0	0	0	0	0.06 $\cong$ 0	65.53	65.12			0.271	0.271	0.0038	0.004	73.454	73.474
t+6	71.28	0	0	0	0	1	1	0	0.55 $\cong$ 1			31.4	31.94	-2.18	-2.157	-0.0296	-0.030	71.297	71.269
t+7	71.95	1	1	1	1	0	0	0	0.35 $\cong$ 0	64.02	63.66			0.669	0.666	0.0094	0.009	71.963	71.911
t+8	71.07	0	0	0	0	1	1	0	0.02 $\cong$ 0			31.61	31.88	-0.88	-0.885	-0.0122	-0.012	71.078	71.048
t+9	73.88	1	1	1	1	0	0	0	0.48 $\cong$ 0	65.95	66.09			2.809	2.821	0.0395	0.040	73.899	73.890
t+10	70.61	0	0	0	0	1	1	1	1.10 $\cong$ 1			32.07	31.16	-3.27	-3.289	-0.0443	-0.044	70.61	70.639
t+11	71.44	1	1	0	0	0	0							0.830	0.835	0.0117	0.012	71.445	71.486
t+12	72.78	1	1	1	1	0	0	2	0.23 $\cong$ 0	64.85	65.45			1.339	1.346	0.0187	0.019	72.791	72.845
t+13	72.33	0	0	0	0	0	0							-0.45	-0.454	-0.00618	-0.006	72.337	72.407
t+14	71.79	0	0	0	0	0	0							-0.54	-0.541	-0.00747	-0.007	71.796	71.901
t+15	71.21	0	0	0	0	1	0	0	0.37 $\cong$ 0			31.47	31.56	-0.58	-0.582	-0.00808	-0.008	71.214	71.325
t+16	71.8	1	1	1	1	0	0	0	0.25 $\cong$ 0	63.87	64.18			0.590	0.592	0.00828	0.008	71.806	71.896
t+17	69.14	0	0	0	0	1	1	0	1.14 $\cong$ 0			33.54	33.50	-2.66	-2.675	-0.03705	-0.037	69.131	69.236
t+18	69.31	1	1	1	0	0	0	0	0.06 $\cong$ 0	61.38	61.99			0.169	0.166	0.00246	0.002	69.297	69.374
t+19	66.78	0	0	0	0	1	1	0	0.62 $\cong$ 1			35.9	36.05	-2.53	-2.535	-0.0365	-0.036	66.762	66.877

4.12. Return prediction

4.12.1. Return prediction using ANN model

The results of stock return prediction using MLP model are presented in Table 26. As can be seen, the prediction model has a good performance for all selected stocks. The R^2 values are greater than 0.9992 for all stocks in train, test, and validation data.

4.12.2. Return prediction using LSTM

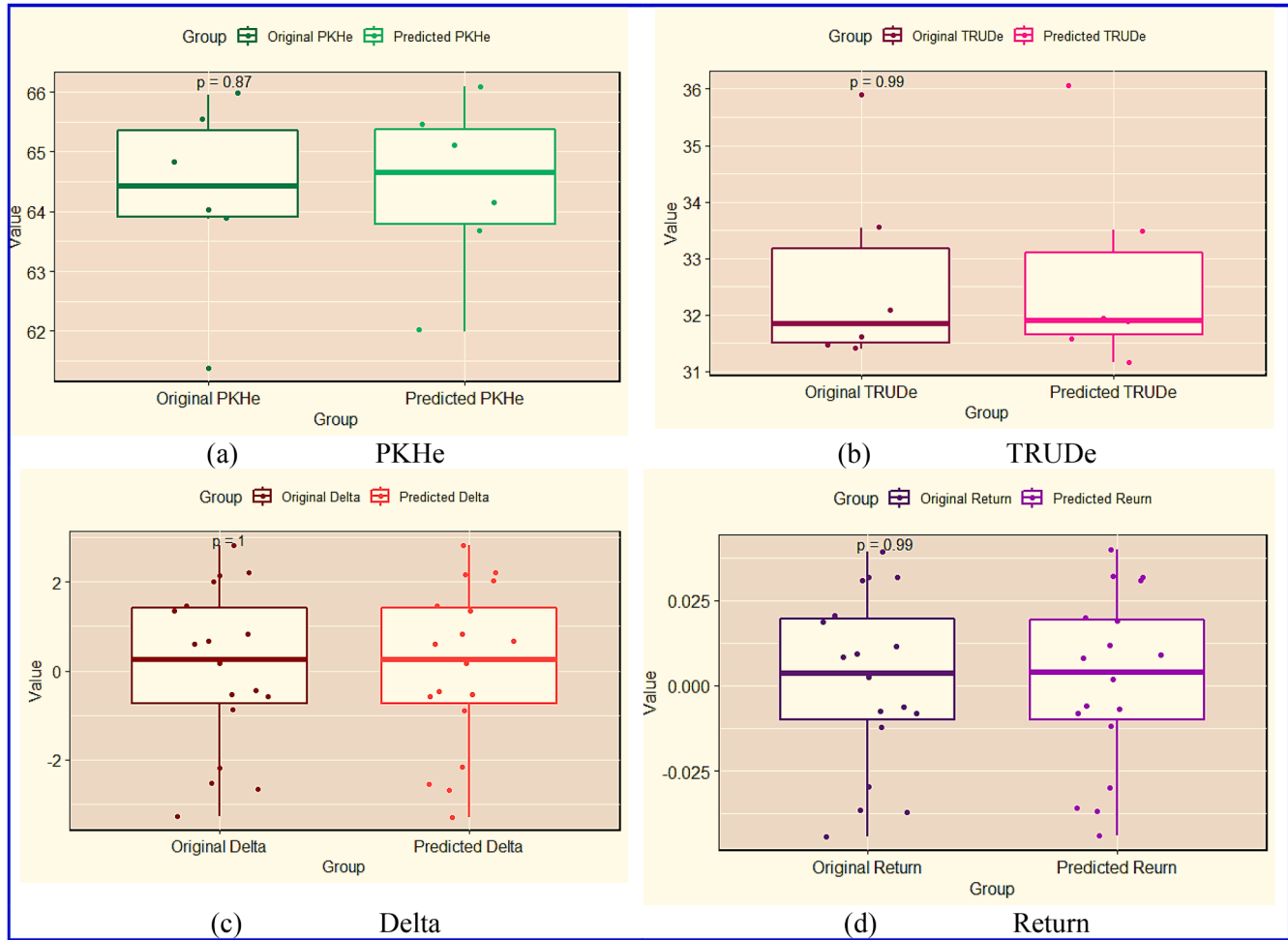
The results of stock return prediction using LSTM are presented in Table 27.

Fig. 12 comparing the real target and the output of proposed ANN for "Return" in four selected stocks.

Table 29Two Sample *t*-test results.

Attribute	t	df	p-value	95 percent confidence interval		mean in group Original	mean in group Predicted
PKHe	-0.16515	9.89	0.8721	-2.152320	1.855653	64.26667	64.41500
TRUDe	-0.016011	9.89	0.9875	-2.336398	2.303065	32.66500	32.68167
Delta	-0.000363	36	0.9997 $\approx$ 1	-1.176272	1.175851	0.07463158	0.07484211
Return	-0.008624	36	0.9932	-0.016657	0.016516	0.001456	0.001526

alternative hypothesis: true difference in means is not equal to 0.

**Fig. 13.** Comparison of original and predicted values with boxplot.**Table 30**Two Sample *t*-test results for comparing original and predicted closing prices.

Attribute	t	df	p-value	95 percent confidence interval		mean in group Original	mean in group Predicted
CP1-CP2	36	0.0062259	0.9951	-1.264825	1.272615	71.08653	71.08263
CP1-CP3	36	0.06944	0.945	-1.220309	1.306835	71.12589	71.08263
CP2-CP3	36	-0.063106	0.95	-1.304587	1.225850	71.08653	71.12589

alternative hypothesis: true difference in means is not equal to 0.

4.12. Validation of the proposed MLP prediction method

In order to check the validity of proposed prediction approach the BHP stock is selected.

The results of the prediction of continuous indices and classification of binary targets of BHP stock for the selected period are presented in subSection 4.2 to 4.11.

Considering that the R2 values for validation data for DBP and DBT

indices of BHP stock were about 34% and 42%, respectively, these indices were removed and validation was performed with the remaining eight indices. The prediction of Engineered and Derivative indices for the next 19 days (January 4, 2021, and January 29, 2021) for BHP stock has been accomplished using the proposed models. The results are presented in Table 28.

As can be seen, the original closing prices of the BHP stock for January 4th to January 29th are presented in the first column of

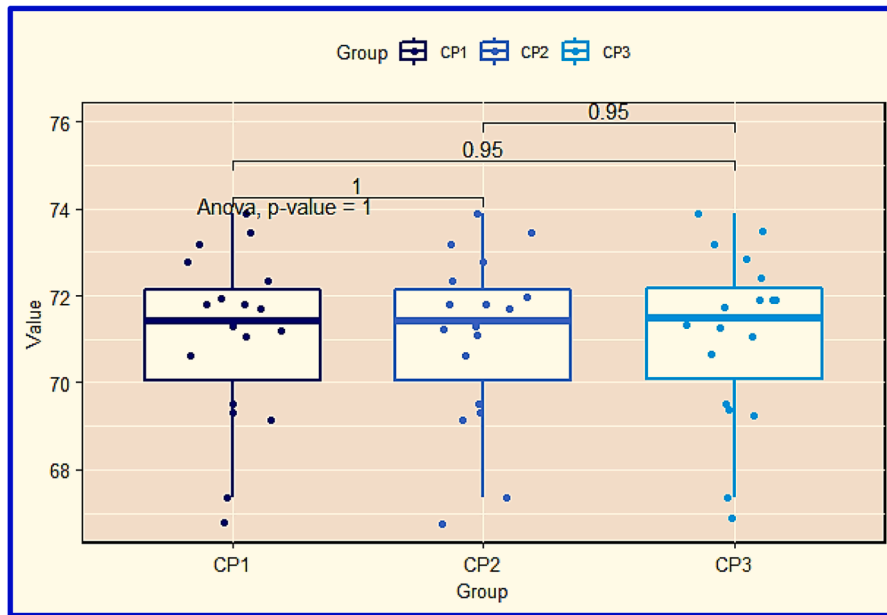


Fig. 14. Comparison of original and predicted values of closing price with boxplot.

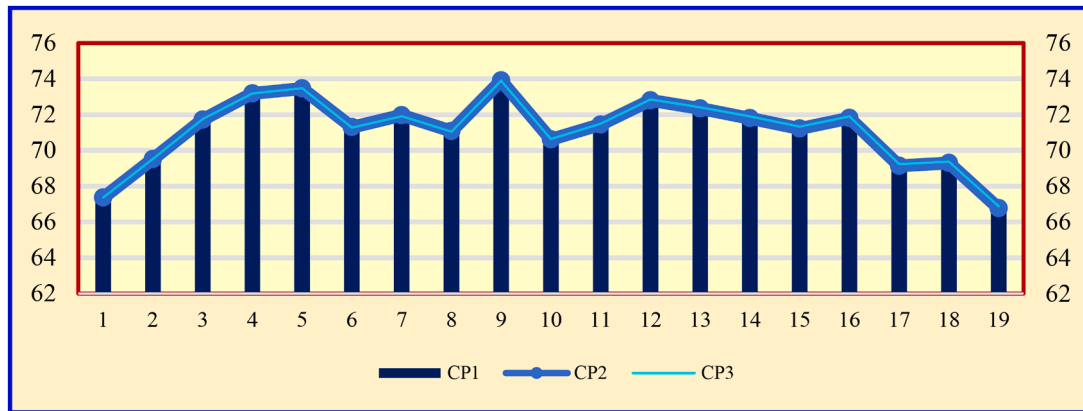


Fig. 15. original and predicted values of closing price for the next 19 days.

Table 28.

Firstly, the performance of ‘Trend’, ‘PK’, and ‘TRU’ indices classification models have been evaluated.

- The original value of the trend index is also presented in the second column. The results of the classification ‘Trend’ is presented in the third column of Table 28. By comparing the original and predicted values of this index, it can be said that the model for the classification of this variable has performed well.
- The original and predicted values of the ‘PK’ target variable are also presented in the fourth and fifth columns of Table 28, respectively. As mentioned in Section 4.3, the prediction model of this target variable has an accuracy of about 75.6%. It can be seen that this target variable has not been correctly predicted for some days.
- The original and predicted values of the ‘TRU’ target variable is presented in the sixth and seventh columns of Table 28, respectively. As can be seen, this variable was not correctly predicted on day $t + 15$ and the prediction for the other days is done correctly and the model accurately predicts the troughs.

In the following, the validity of continuous indices forecasting models including ‘DBPT’, ‘PKHe’, ‘TRUde’, ‘Delta’, and ‘Return’, is

discussed.

- The original and predicted value of the DBPT index are presented in the Eighth and ninth columns of Table 16. According to the original value of PK index (column 4), The price is at the peak on day $t + 5$. the distance from this peak to the next trough is equal to zero. The prediction value for the distance to the next peak is equal to 0.06 and therefore the prediction model had an acceptable performance on this day. Predictions for $t + 12$, $t + 17$, and $t + 19$ were not accurate. As mentioned in Section 4.7, the variable prediction model of this target variable has a coefficient of determination of about 78.78%.

The original and predicted values of ‘PKHe’, ‘TRUde’, ‘Delta’, and ‘Return’ indices are presented in columns 10 to 17 of Table 28. To determine whether there is any significant difference between the average of original values and predicted values of these indices, independent t -test have been used, and the test hypothesis considered as Eq. (27)

$$\begin{cases} H_0 : \mu_{\text{Original}} = \mu_{\text{Predicted}} \\ H_1 : \mu_{\text{Original}} \neq \mu_{\text{Predicted}} \end{cases} \quad (27)$$

If the p -value in the output of the independent t -test is lower than the

Type I error, the null hypothesis is rejected, meaning that there is a significant relationship between the means. Also, if one of the upper bound and the lower bound is positive and the other is negative, this means that the means of the two groups are not significantly different. The results of the t -test of each of these indices are presented in Table 29.

As can be seen in Table 29, in all cases the p -value is greater than the Type I error, and also the lower bound is negative and the upper bound is positive, therefore the means of the two groups for each mentioned index are not significantly different.

Fig. 13 presents the comparison of original and predicted values of these indices with a boxplot.

In the following, the closing price for the selected period (January 4, 2021, and January 29, 2021) have been calculated using the predicted values of 'Delta' and 'Return'.

Considering the availability of the closing price on day t (65.34) and using the predicted value of the 'Delta' target variable (column 15) and Eq. (9), the closing price on day $t + 1$ can be calculated as Eq. (28):

$$\hat{P}_{t+1} = \hat{\Delta}_{t+1} + P_t = 2.017 + 65.34 = 67.357 \quad (28)$$

The closing price on day $t + 2$ can also be calculated according to the price calculated for day $t + 1$ using Eq. (9).

$$\hat{P}_{t+2} = \hat{\Delta}_{t+2} + \hat{P}_{t+1} = 2.155 + 67.35 = 69.512 \quad (29)$$

The closing price for the coming days is calculated in the same way. The calculated closing price using predicted 'Delta' (CP2) are presented in column 18 of Table 28.

The predicted value of the 'Return' is presented in the column 17 of Table 28. Once again, this time using Eq. (10), the closing price on day $t + 1$ can be calculated as Eq. (30):

$$\hat{P}_{t+1} = \widehat{Ret}_{t+1} \times P_t + P_t = 0.031 \times 65.34 + 65.34 = 67.366 \quad (30)$$

The calculated Closing price using predicted 'Return' (CP3) are presented in column 19 of Table 28.

In order to compare the average of original closing price (CP1), calculated closing price using predicted 'Delta' (CP2), and calculated closing price using predicted 'Return' (CP3), independent t -test have been used, and the test hypothesis considered as Eq. (31).

$$\begin{cases} H_0 : \mu_{CP1} = \mu_{CP2} = \mu_{CP3} \\ H_1 : \text{At least between two groups, the averages are different} \end{cases} \quad (31)$$

The results of the average of CP1, CP2, and CP3 are presented in Table 30. Fig. 14 presents the comparison of the original and predicted closing prices using a boxplot. As can be seen, there is no significant difference between the means according to P -values. CP1, CP2, and CP3 are provided in Fig. 15 for the next 19 days. It is clear that there are similarities.

5. Conclusion remarks and suggestion for future researches

Accurate estimation of future stock prices can be of significant benefit to investors in the field of stock exchanges. In this paper, machine learning approach designed to predict the next day's stock prices using artificial neural networks (ANNs), with a particular focus on engineered and derivative indices. By extracting a novel set of indices from historical stock price data, this research enhances predictive capabilities and provides targeted insights that traditional forecasting methods often overlook. The proposed model not only predicts well-established indicators such as price trends and returns but also introduces innovative metrics, including peak and trough predictions and distance measurements between significant price points.

The proposed approach had two main stages. In the first stage, 5283 records of daily historical series of open prices, close prices, the highest prices, the lowest prices, and the trading volume of assets of four well-known stocks were selected from FTSE 100. Afterward, a novel set of engineered and derivative indices extracted from the original time series

for stock price prediction were proposed. In the second stage, two Multi-Layer Perceptron (MLP) were proposed to predict the next day's stock prices using the engineered discrete and continuous indices.

The proposed approach in this paper helps the investor in predicting the stock price by presenting a set of new indices extracted from the closing price of stocks. There are several one-day-ahead targets that can be interesting for investors. For example, forecasting the amount of increase of the closing price in the next day, the return of the next day, the price trend in the next day. In addition to these well-known indicators, some novel indices such as determining whether the price will be at its peak the next day or not? Predicting whether the price will be in the trough the next day or not? predicting the height of the next peak, predicting the depth of the next trough, predicting the distance from one peak to the next peak, distance from one trough to the next trough, the distance of one peak from the next trough and vice versa, were proposed and developed in this research. These novel indices can be very attractive for the investor. We called such indicators engineered and derivative indices. The aforementioned indices were extracted from daily stock price data to be considered as target variables. In fact, in this research, instead of directly predicting stock prices, engineered indices were used. This study employs engineered and derivative indices because they can enhance predictive capability, provide targeted insights, and improve interpretability while addressing the limitations of traditional forecasting methods. By leveraging these features, the research aims to contribute to developing more robust and reliable stock price prediction models that can better serve investors in navigating the complexities of financial markets.

Ten different day-ahead indices were defined to predict these indices. A set of input variables, including several lagged variables of returns and several technical indicators, were considered. The proposed approach is implemented on four stocks, namely "BP," "BHP," "GSK," and "VOD," that were selected from the daily stock exchange of FTSE 100 between January 1, 2000, and December 31, 2020. The case study involving well-known stocks from the FTSE 100 demonstrates the practical applicability of the proposed model. Investors can leverage these findings to enhance their trading strategies in actual market conditions. Results revealed that by integrating various predictive factors and innovative target variables, investors can gain valuable insights into potential price movements.

One of the key innovations of the proposed model is its use of a comprehensive set of engineered and derivative indices extracted from daily stock price time series data. This approach allows for a broader range of input variables, including predictions of trends, peak and trough detection, and distances between significant price points, which have not been previously explored in the literature. By leveraging these indices, the model captures complex and hidden patterns in stock price movements, potentially leading to more accurate predictions than reliance on past daily returns. In terms of usability, the proposed approach allows investors to make more informed decisions, ultimately improving their investment strategies. The implications of this research also extend to businesses. Companies can enhance their market strategies and financial planning by analyzing stock price trends and the factors that drive them. For instance, the insights derived from the proposed model can help organizations determine the optimal timing for stock issuances, effectively manage their capital structures, and refine their investor relations strategies. Financial analysts and portfolio managers can leverage the model's predictions to identify undervalued stocks with significant growth potential, enabling them to build portfolios more likely to outperform the market.

Additionally, by anticipating fluctuations in stock prices, they can proactively adjust their portfolios to reduce risks and safeguard investors' capital. The model's focus on multiple prediction targets provides users with richer insights, enabling better-informed trading strategies. This contrasts with several mentioned in previous research, which primarily focus on predicting next-day returns, potentially limiting its applicability to broader investment strategies.

This study has several limitations that should be acknowledged. The model primarily relies on engineered and derivative indices derived from historical stock price data, which may restrict its predictive capabilities by not incorporating fundamental variables that could enhance accuracy. Secondly, the analysis is based on a limited dataset of four stocks from the FTSE 100, which may affect the generalizability of the findings to other markets or a broader range of stocks.

The suggested approach can be improved in the following aspects in future research. Some fundamental variables can also be considered as the input variables. Also, some base indicators such as gold, oil, gas prices, interest and inflation rates, import and export rates, and GDP may be considered in future research. Extending the prediction window to longer time frames, such as weekly or monthly, could enhance the practical applicability of stock price prediction approaches for investors with different time horizons. Focusing solely on next-day stock prices limits the usefulness of these models to short-term traders. By developing models capable of forecasting stock prices over weekly or monthly intervals, the research would cater to a broader range of investors, including those with medium-term and long-term investment strategies. Expanding the prediction horizon would make the findings more widely applicable and valuable to the investment community. The proposed approach in this research can be used to preselect stocks to form a portfolio. Therefore, combining this approach with portfolio optimization problems may be engaging in future research.

CRedit author statement

Shaghayegh Abolmakarem: PhD Student: Modeling, solution procedures, Data gathering, Software, Writing- Original draft preparation

Farshid Abdi: PhD, Assistant Professor, Supervisor: data mining, modeling, solution procedures, commenting on total structure of the manuscript.

Kaveh Khalili-Damghani: PhD, Associate Professor, Co-Supervisor: Conceptualization, Modeling, Solution Procedures; Analyzing the results, Writing.

Hosein Didehkhani: PhD, Assistant Professor, Adviser: Portfolio Optimization, Investment Market, commenting on total structure of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.iswa.2024.200449](https://doi.org/10.1016/j.iswa.2024.200449).

References

- Abe, M., & Nakayama, H. (2018). Deep learning for forecasting stock returns in the cross-section. *Advances in knowledge discovery and data mining* (pp. 273–284). Springer International Publishing. pp.
- Ampomah, E. K., Qin, Z., Nyame, G., & Botchey, F. E. (2020). Stock market decision support modeling with tree-based adaboost ensemble machine learning models. *Informatica*, 44, 477–489.
- Ayyildiz, N., & Iskenderoglu, O. (2024). How effective is machine learning in stock market predictions? *Heliyon*, 10(2). <https://doi.org/10.1016/j.heliyon.2024.e24123>
- Ayyildiz, N. (2023). Prediction of stock market index movements with machine learning. *Özgür Publications*. <https://doi.org/10.58830/ozgur.pub354>
- Ayyildiz, N., & Iskenderoglu, O. (2023). Prediction of stock index movement directions using machine learning methods: An application on developing countries. *Journal of Financial Economics and Banking*, 4(2), 68–78.
- Baek, Y., & Kim, H. Y. (2018). ModAugNet: A new forecasting framework for stock market index value with an overfitting prevention LSTM module and a prediction LSTM module. *Expert Systems with Applications*, 113, 457–480.
- Ballings, M., Van Den Poel, D., Hespeels, N., & Gryp, R. (2015). Evaluating multiple classifiers for stock price direction prediction. *Expert Systems with Applications*, 42 (20), 7046–7056.
- Basak, S., Kar, S., Saha, S., Khaidem, L., & Roy Dey, S. (2019). Predicting the direction of stock market prices using tree-based classifiers. *The North American Journal of Economics and Finance*, 47, 552–567.
- Bisoi, R., Dash, P. K., & Parida, A. K. (2019). Hybrid Variational Mode Decomposition and evolutionary robust kernel extreme learning machine for stock price and movement prediction on daily basis. *Applied Soft Computing*, 74, 652–678.
- Boltona, J., & Boettichera, S. (2015). Momentum trading on the Johannesburg stock exchange after the global financial crisis. *Procedia Economics and Finance*, 24(2015), 83–92.
- Bruni, R. (2017). Stock market index data and indicators for day trading as a binary classification problem. *Data in Brief*, 10, 569–575.
- Chen, Y., & Hao, Y. (2018). Integrating principle component analysis and weighted support vector machine for stock trading signals prediction. *Neurocomputing*, 321, 381–402.
- Chen, W., Zhang, H., Mehlatat, M. K., & Jia, L. (2021). Mean-variance portfolio optimization using machine learning-based stock price prediction. *Applied Soft Computing*, 100, Article 106943.
- Chen, K., Zhou, Y., & Dai, F. (2015). A lstm-based method for stock returns prediction: A case study of china stock market. In *2015 IEEE International Conference on Big Data (Big Data)*. IEEE.
- Dai, Z., Zhu, H., & Kang, J. (2021). New technical indicators and stock returns predictability. *International Review of Economics and Finance*, 71, 127–142.
- Dezsi, E., & Nistor, I. A. (2016). Can deep machine learning outsmart the market? a comparison between econometric modelling and long- short term memory. *Romanian Economic Business Review*, 11(4.1), 54–73.
- de Almeida, R., Reynoso-Meza, G., & Arns Steiner, M. T. (2016). Multi-objective optimization approach to stock market technical indicators. In *2016 IEEE Congress on Evolutionary Computation (CEC)*, Vancouver, BC, Canada (pp. 3670–3677). <https://doi.org/10.1109/CEC.2016.7744254>
- Erzurum Cicek, Z. I., & Kamisli Ozturk, Z. (2021). Optimizing the artificial neural network parameters using a biased random key genetic algorithm for time series forecasting. *Applied Soft Computing*, 102, 107091. <https://www.sciencedirect.com/science/article/abs/pii/S1568494621000144>.
- Freitas, F. D., Souza, A. F. D., & Almeida, A. R. D. (2009). Prediction-based portfolio optimization model using neural networks. *Neurocomputing*, 72, 2155–2170.
- Fu, X., Du, J., Guo, Y., Liu, M., Dong, T., & Duan, X. (2018). A machine learning framework for stock selection. <https://doi.org/10.48550/arXiv.1806.01743>.
- Gandhmal, D. P., & Kumar, K. (2019). Systematic analysis and review of stock market prediction techniques. *Computer Science Review*, 34, Article 100190.
- Gorgulho, A., Neves, R., & Horta, N. (2011). Applying a GA kernel on optimizing technical analysis rules for stock picking and portfolio composition. *Expert Systems with Applications*, 38, 14072–14085.
- Hiransha, M., Gopalakrishnan, E. A., Menon, V. K., & Soman, K. P. (2018). NSE stock market prediction using deep-learning models. *Procedia Computer Science*, 132, 1351–1362.
- Huang, W., Lai, K. K., Nakamori, Y., & Wang, S. (2004). Forecasting foreign exchange rates with artificial neural networks: A review. *International Journal of Information Technology and Decision Making*, 3(1), 145–165.
- Ismail, M. S., Md Noorani, M. S., Ismail, M., Abdul Razak, F., & Alias, M. A. (2020). Predicting next day direction of stock price movement using machine learning methods with persistent homology: Evidence from Kuala Lumpur Stock Exchange. *Applied Soft Computing*, 93, Article 106422.
- Kara, Y., Acar Boyacioglu, M., & Baykan, O. K. (2011). Predicting direction of stock price index movement using artificial neural networks and support vector machines: The sample of the Istanbul Stock Exchange. *Expert Systems with Applications*, 38(5), 5311–5319.
- Kumar Chandar, S., Sumathi, M., & Sivanandam, S. N. (2016). Prediction of stock market price using hybrid of wavelet transform and artificial neural network. *Indian Journal of Science and Technology*, 9(8), 1–5. <https://doi.org/10.17485/ijst/2016/v9i8/87905>
- Kumar Chandar, S. (2021). Hybrid models for intraday stock price forecasting based on artificial neural networks and metaheuristic algorithms. *Pattern Recognition Letters*, 147, 124–133.
- Lahmiri, S. (2018). Minute-ahead stock price forecasting based on singular spectrum analysis and support vector regression. *Applied Mathematics and Computation*, 320, 444–451.
- Lee, M.-C. (2009). Using support vector machine with a hybrid feature selection method to the stock trend prediction. *Expert Systems with Applications*, 36, 10896–10904.
- Lee, S. I., & Yoo, S. J. (2020). Threshold-based portfolio: The role of the threshold and its applications. *The Journal of Supercomputing*, 76, 8040–8057. <https://doi.org/10.1007/s11227-018-2577-1>
- Lei, L. (2018). Wavelet neural network prediction method of stock price trend based on rough set attribute reduction. *Applied Soft Computing*, 62, 923–932.
- Long, W., Lu, Z., & Cui, L. (2019). Deep learning-based feature engineering for stock price movement prediction. *Knowledge-Based Systems*, 164, 163–173.

- Luque, A., Carrasco, A., Martfina, A., & de las Heras, A. (2019). The impact of class imbalance in classification performance metrics based on the binary confusion matrix. *Pattern Recognition*, 91, 216–231.
- Ma, Y., Han, R., & Wang, W. (2021). Portfolio optimization with return prediction using deep learning and machine learning. *Expert Systems with Applications*, 165, Article 113973.
- Mansini, R., Ogryczak, W., & Speranza, M. G. (2015). *Linear and mixed integer programming for portfolio optimization*. Springer.
- Murtagh, F. (1991). Multilayer perceptrons for classification and regression. *Neurocomputing*, 2(5–6), 183–197.
- Patel, J., Shah, S., Thakkar, P., & Kotecha, K. (2015). Predicting stock market index using fusion of machine learning techniques. *Expert Systems with Applications*, 42(4), 2162–2172.
- Paiva, F. D., Cardoso, R. T. N., Hanaoka, G. P., & Duarte, W. M. (2019). Decision-making for financial trading: A fusion approach of machine learning and portfolio selection. *Expert Systems with Applications*, 115, 635–655.
- Pawar, K., Jalem, R.S., & Tiwari, V. (2019). Stock market price prediction using LSTM RNN. In: V.Rathore, M.Worring, D.Mishra, A.Joshi, & S.Maheshwari (eds) *Emerging trends in expert applications and security. advances in intelligent systems and computing*, vol. 841. Springer, Singapore. https://doi.org/10.1007/978-981-13-2285-3_58.
- Ribeiro, G. T., Santos, A. A. P., Mariani, V. C., & Coelho, L dos Santos (2021). Novel hybrid model based on echo state neural network applied to the prediction of stock price return volatility. *Expert Systems with Applications*, 184, Article 115490.
- Siswanto, J., Prabuwo, A. S., Abdullah, A., & Idrus, B. (2016). A linear model based on Kalman filter for improving neural network classification performance. *Expert Systems with Applications*, 49, 112–122.
- Subha, M. V., & Nambi, S. T. (2012). Classification of stock index movement using k-nearest neighbours (k-NN) algorithm. *WSEAS Transactions on Information Science and Applications*, 9(9), 261–270.
- Tan, Z., Yan, Z., & Zhu, G. (2019). Stock selection with random forest: An exploitation of excess return in the Chinese stock market. *Heliyon*, 5(8), e02310.
- Taran-Moroşan, A. (2011). The relative strength index revisited. *African Journal of Business Management*, 5(14), 5855–5862.
- Verma, S., Verma, S., & Prasad Sahu, T. (2023). Stock Market Forecasting with Different Input Indicators using Machine Learning and Deep Learning Techniques: A Review. *Engineering Letters*, 31(1), 213–229.
- Vijh, M., Chandola, D., Tikkiwal, V. A., & Kumar, A. (2020). Stock closing price prediction using machine learning techniques. *Procedia Computer Science*, 167, 599–606.
- Wang, W., Li, W., Zhang, N., & Liu, K. (2020). Portfolio formation with preselection using deep learning from long-term financial data. *Expert Systems with Applications*, 143, Article 113042.
- Weng, B., Ahmed, M. A., & Megahed, F. M. (2017). Stock market one-day ahead movement prediction using disparate data sources. *Expert Systems with Applications*, 79, 153–163.
- Wilder, J. W. (1978). New concepts in technical trading systems. *Trend Research*.
- Xu, Y., Chhim, L., Zheng, B., & Nojima, Y. (2020). Stacked deep learning structure with bidirectional long-short term memory for stock market prediction. In H. Zhang, Z. Zhang, Z. Wu, & T. Hao (Eds.), *Neural computing for advanced applications. NCA 2020. communications in computer and information science*. Singapore: Springer. vol 1265.
- Yang, F., Chen, Z., Li, J., & Tang, L. (2019). A novel hybrid stock selection method with stock prediction. *Applied Soft Computing*, 80, 820–831.
- Yao, J., Tan, C., & POH, H.-L. (1999). Neural Networks for Technical Analysis: A Study on KLCI. *International Journal of Theoretical and Applied Finance*, 2(2), 221–241.
- Zhang, Y., Yan, B., & Aasma, M. (2020). A novel deep learning framework: Prediction and analysis of financial time series using CEEMD and LSTM. *Expert Systems with Applications*, 159, Article 113609.
- Zhou, F., Zhou, H.-M., Yang, Z., & Yang, L. (2019). EMD2FNN: A strategy combining empirical mode decomposition and factorization machine based neural network for stock market trend prediction. *Expert Systems with Applications*, 115, 136–151.